ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2022)4072706 - 01/06/2022

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101061729
Project name:	REshaping Supply CHAins for Positive social impact
Project acronym:	ReSChape
Call:	HORIZON-CL2-2021-TRANSFORMATIONS-01
Topic:	HORIZON-CL2-2021-TRANSFORMATIONS-01-06
Type of action:	HORIZON-RIA
Service:	REA/C/01
Project starting date:	fixed date: 1 October 2022
Project duration:	36 months

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PROJECT SUMMARY

Project summary *Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.* *Use the project summary from your proposal.*

The aims of this proposal are:

- To analyse social, economic and environmental changes and disruptions (including covid) and evaluate their impacton SCs, identifying related challenges in terms of relationship between countries, configuration of the network, impact on employment.
- To study and propose a set of SC models for the evolution of global SC integrating strategies like resource efficient,closed-loop and humanitarian as a way to increase EU resilience and sustainability. Particular attention will be given to the role of digitalization as a way to establish new paths for social inclusion taking into consideration the needs of urban and rural areas. Some important European sectors like fashion, automotive, medical and machine tools will be analysed with case studies and survey.
- To develop Innovative tools for monitoring and assessing sectoral trade patterns and defining mechanisms to evaluaterelationship of disruptions like pandemic and global value chains taking into consideration impact on employment, economic growth, incomes etc also in the long term. Moreover, it will be analysed the impact of different trade patterns, on the EU value added of sectoral and countries with a specific focus on analyzing income inequalities and proposition of decent work and social cohesion. Particular attention will be gives to gender issues and social disparities.
- To develop innovative policy scenarios with recommendations for future global value chains: policy scenarios will bebased on Key horizontal issues impacting on several sectors and will provide recommendations for EU, national and sectoral strategies, policy measures and targeted actions aimed at shaping fair, inclusive and sustainable trade patterns, value and supply chains as well as production networks.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	CNR	CONSIGLIO NAZIONALE DELLE RICERCHE	IT	999979500
1.1	AE	UNIPD	UNIVERSITA DEGLI STUDI DI PADOVA	IT	999995602
2	BEN	INESC TEC	INESC TEC - INSTITUTO DE ENGENHARIADE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIENCIA	PT	999513706
3	BEN	Fraunhofer	FRAUNHOFER GESELLSCHAFT ZUR FORDERUNG DER ANGEWANDTEN FORSCHUNG EV	DE	999984059
4	BEN	ZLC	FUNDACION ZARAGOZA LOGISTICS CENTER	ES	999608960
5	BEN	RWI	RWI - LEIBNIZ-INSTITUT FUR WIRTSCHAFTSFORSCHUNG e.V.	DE	972715807
6	BEN	UDEUSTO	UNIVERSIDAD DE LA IGLESIA DE DEUSTO ENTIDAD RELIGIOSA	ES	999876971
7	BEN	TU/e	TECHNISCHE UNIVERSITEIT EINDHOVEN	NL	999977269

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC

8	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
9	AP	ASTON U	ASTON UNIVERSITY	UK	999896953

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LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverable No(s)
WP1	Project Coordination	1 - CNR		1	36	D1.3, D1.2, D1.1
WP2	Global trends and Innovative supply chain models	1 - CNR		1	18	D2.1, D2.2, D2.3, D2.4
WP3	Impact of new SC models on social indicators	9 - ASTON U		10	33	D3.6, D3.1, D3.5, D3.3, D3.4, D3.2
WP4	Development of policy scenarios	2 - INESC TEC		18	36	D4.2, D4.1, D4.3
WP5	Dissemination and Communication	4 - ZLC		1	36	D5.1, D5.3, D5.2

Work package WP1 – Project Coordination

Work Package Number	WP1	**Lead Beneficiary**	1. CNR
Work Package Name	Project Coordination		
Start Month	1	**End Month**	36

Objectives

Objectives
To efficiently manage and administrate all project internal and external activities. Therefore, an adequate management structure is necessary to handle coordination of work carried out in project, quality control, financial and thematic reporting and any communication among the project partners, the associated members and the stakeholders in general.
Specific objectives are:

- Manage the communication in the Project Consortium and Decision Board
- Give overall project management and provide for quality management ensuring the timely achievement of the projectgoals as outlined in the work plan.
- To interface with the European Commission and other partners.
- Ensure coherence with all legal and EC requirements, rules and regulations and supervise fulfilment of ConsortiumAgreement, including financial and legal management.
- Manage the “Expert Group” and generally the stakeholders to be involved.

Description

T1.1 Consortium and project management (Task leader: CNR; Participants: All) (M01-M36)
● Ensure that received funds are correctly distributed and accounted for.
● Establish a functional management structure for fruitful interaction among the partners, efficient communication andsmooth reporting to the EC, e.g. by implementing project management, work plan, financial and activity reporting tools.
● Provide assistance and advice to the project partners regarding administration and reporting.
● Monitor budget and expenditure as well as effective and efficient execution of contract and work plan.● Represent the consortium towards the European Commission and other institutions, ● Organise regular meetings in the Project Consortium and Decision Board.
● Elaborate periodic management and progress reports to regularly inform the European Commission on the activitiesof the project (work progress, financial statements).
● Moreover, the project coordinator will manage the technical and scientific coordination of the project. In particular,it will manage the interrelation between the WPs of the project:
● Monitor the technical advances of the project in each work packages through a close interaction with all work packageleaders.
● Ensure that the project advances as according to the work plan of the project.
● Observe external developments and take action for strategic adaptation of the project.

T1.2-Quality management (Task leader: CNR; Participants: All) (M01-M36)
The quality management of the project consists in continuously checking that results and achievements are in line with the objectives fixed and to ensure that the project orientations, activities and outcomes are efficiently answering the needs of the European manufacturing community in the process industry. Also, the objective is to ensure a timely completion of all deliverables as according to the work plan and provide quality information. The task leader will be responsible for observing the review processes and to propose actions to counter undesirable developments. These will be discussed within the consortium and the follow-up will be ensured by the task leader and the coordinator. All partners will be incorporated into the reviewing and monitoring work flow within the respective work packages. The work package leaders will have a more pronounced role.

T1.3-Data Management Plan (Task leader: ZLC; Participants: CNR) (M01-M36)
DMPs will include information about data handle, managed, and retrieved by methods, tools, techniques used in the project. It will also include a guideline for ethics compliance (for personal data gathering). The DMPs will establish the policy regulating collection, management, sharing, archiving, and preservation of outcome data in the ReSChape project providing the identification elements and the descriptions of the data sets. As part of making data and metadata FAIR, information will include: (a) how data will be handled, curated and preserved (during and after the end of the project); (b) what data will be collected, processed and/or generated; (c) which methodology & standards will be applied; (d) data open access procedure; (e) recommendations on fulfilment of FAIR requirements for metadata flows.

T1.4. Ethics compliance (Task leader: CNR; Participants: All) (M01-M36) In this task, the consortium will take care of possible issues related to ethics compliance which could be mainly related to the involvement of experts from different sectors, their participation to workshops, focus groups, and other interactive activities. It is clear that ReSChape project activities are mainly related to analyse aggregate data and no personal data are necessary or of interest for the aim of our studies.

Work package WP2 – Global trends and Innovative supply chain models

Work Package Number	WP2	**Lead Beneficiary**	1. CNR
Work Package Name	Global trends and Innovative supply chain models		
Start Month	1	**End Month**	18
Objectives			

Objectives
The goal of this WP is to propose innovative models for SCs shaped by various socio-economic, political and technological trends and changes. In doing so, challenges and opportunities will be outlined, which will help to define how SCs strategies and processes can be improved to make them more social oriented.
Taking into consideration priorities of social resilience and sustainability, special insight will be given to analyse how digital technologies can support this process. The approach applied will be based both on the analysis of market needs and on the technologies available and to be developed to define paths of innovation.

Description

Description of work
T2.1-Identification of trends, changes and disruptions (Task leader: IML; Participants: CNR , INESC TEC, ZLC, RWI, TUE) (M01-M06)

- Performing a PESTLE analysis to investigate trends, changes, and disruptions with economical, ecological, and/orsocial impacts on European SCs
- Examining NEXT-NET findings with regard to their influence on resilience and sustainability
- Conducting a structured literature analysis to ensure the inclusion of newest insights into the state of affairs
- Supplementation and validation of the results together with various stakeholders● Written report on the results

T2.2- Social and sustainability challenges for SC of the future (Task leader: CNR; Participants: INESC TEC, ZLC) (M04-M08)
This task requires input from T2.1, as trends and changes affecting the design and operations of global SCs will reveal what challenges will arise. Challenges will be classified according to dimensions like behavioural, operational, economic and technological. Different trends may cause similar challenges and opportunities. In addition, the relationships among different challenges further complicate the analysis. In doing so, we will take the complex interactions and dependencies amongst several trends and challenges/opportunities into account. This will be based on a preliminary analysis made by the partners to transform trends and changes in challenges and will be validated with focus groups involving experts from different industrial sectors. Moreover, challenges and opportunities will be categorised based on social and sustainability dimensions.

T2.3-Definition of SC strategies for resilience and sustainability (Task leader: INESC TEC; Participants: CNR, ASTON, IML, ZLC, RWI, DEUSTO) (M06-M12)
Considering the social and sustainability challenges for SCs of the future resulting from T2.2, in combination with the trends, changes and disruptions regarding sustainability aspects and resilience requirements from T2.1, this task has the goal to define the SC strategies needed for achieving resilient and sustainable SCs. To achieve this, the methodological approach consists of a thorough review of the state of the art regarding SC resilience and sustainability, which compromise, amidst others, strategies related to resource efficiency and humanitarian aspects. Afterwards, the SC strategies are defined according to the scenario transfer phase of the scenario building approach (Gausemeier et al., 1998), where effects of the identified challenges, trends and disruptions within the SCs concerning sustainability and resilience are analyzed. To this end, we apply the Consequence Matrix (Barros et al. 2021) where these characteristics and aspects are valued according to their importance and plausibility. Through content analysis and comparison with the literature, it is possible to arrive on a set of SC strategies of the future for resilient and sustainable SCs, such as: green

SCs, closed-loop SCs, resource-efficient SCs, social-responsible SCs and humanitarian SCs, amidst others. Particular attention will be given to possible strategies for EU self-reliance and how problems of energy provision can have a direct or indirect impact on SC strategies also in terms of social impact. The validation of the identified set of SC strategies will be done through expert elicitation within workshops, where participants will range from academic researchers, industrial practitioners to policy makers.

T2.4-Innovative ways to organise SCs (Task leader: CNR; Participants: INESC TEC, ASTON, IML, ZLC, RWI, TUE) (M08-M16)
Once the SC strategies are identified it is necessary to understand how they should be developed in terms of processes and relationship with actors. It will be necessary to devise the characteristics and requirements for each SC strategies based on decision fields. The decision fields can be derived from expert elicitation and literature review, along categories like Supplying, Production and distribution. Based on workshops with experts and focus groups, it will be possible to identify how companies plan to change these processes to make them more social resilient and sustainable. The role of the digital technologies will be studied as a enabler of a new social inclusion (i.e. supporting safety, security, monitoring

working conditions). An insight on the threads of digital technologies will allow to consider also their potential risks and new ways to solve them to avoid violation of workers privacy. The output will represent a list of social indicators that will be measured for each model in the WP4.

Objectives
The main element of this WP is to assess the effects of the trends identified in the previous WPs and the resulting changes in global SCs and trade patterns on socio-economic outcomes, such as GDP, employment, job quality, wages, and income inequality. For this purpose, different types of modelling are applied to assess different types of impact both at macro and micro level. Most specifically, two models will explore the social impacts of specific strategies in SCs i.e., SC risk management strategies and outsourcing, whereas the other two will take a more holistic view by focusing on overall changes in trade and SCs to determine socio-economic impacts. The proposed holistic and granular models that will be developed in this WP will increase transparency leading to more actionable insights and a better understanding of the social impacts, thus helping identify root cause drivers.
Description

Description of work T3.1-Assess impact of covid on supply chain and social indicators (Task leader: ASTON; Participants: CNR, INESC TEC, TUE, IRIS) (M10-M16)
T2.5-Skill and employment in logistics and supply chain (Task leader: ASTON; Participants: CNR) (M12-M18) Based on the T2.2 a great challenge recognised in SCs is the skills shortage. The aim of this task is to understand which are the most relevant skills to be developed for the future SCs to assure a positive impact on employment in terms of e.g. minimising poverty and income inequality. In order to recognise the future skills needed the output of the T2.4 will support this task. In particular, this task will perform a skills gap analysis to identify the gaps between the current skills in the logistics/SC sector and the future required skills for supporting 21st century SCs. The first phase of this task will involve a comprehensive knowledge review and identification of existing/current (AS IS) SC management and logistics skills. Then the next step would be to understand the desired state (TO BE) of the skills which will be informed by the different scenarios and megatrends and also by conducting a content analysis of job advertisements. Based on the most critical skills that would be identified in the previous stage, a questionnaire will be created and distributed to respective employers. This will help to collect the perspectives of business employers on the importance of specific skills competencies so as to provide a holistic understanding of the level and nature of employer demand for specific skills and determine the ability of the labour market to meet this demand, particularly in relation to applicants having the skills and qualifications employers require. Then gaps will be identified, and certain challenges and implications will be discussed with regards to social aspects. The task will conclude with a set of recommendations regarding the most important employable soft and hard skills to reduce unemployment rates with equitable opportunities for all.

Work package WP3 – Impact of new SC models on social indicators

Work Package Number	WP3	**Lead Beneficiary**	9. ASTON U
Work Package Name	Impact of new SC models on social indicators		
Start Month	10	**End Month**	33

The aim of this task is to investigate the effects of COVID-19 on SCs and social performance. More specifically a literature review will be conducted to understand the challenges faced by SCs and logistics during pandemic as well as the mitigation strategies put in place to overcome these issues. Data will be collected from a variety of relevant sources such (e.g. journal articles, expert reports, government and relevant stakeholders' policy documents, published expert interviews and policy feedback literature. Based on the initial phase a framework will be developed which will draw on SC risk management strategies and performance. Performance outcomes in this study mainly refer to the social outcomes (environmental and economic impacts will also be assessed) when implementing different SC actions. Social performance entails indicators such as diversity, excessive working time, child/forced labour. The next step will be to develop a structural equation model to analyse the impact of these SC risk management strategies on the social performance of manufacturing companies. A questionnaire will be developed that will entail the metrics of each construct in the model. The survey method will be adopted to collect the data and then test the hypotheses proposed in this study. The specific steps to apply the PLS-SEM technique are: (1) the specification of the conceptual structural model and hypotheses to be tested, (2) data collection, (3) model validation and (4) estimation and evaluation of the results.

T3.2-Model for evaluating impact of new SCs on social indicators (Task leader: RWI; Participants: CNR, INESC TEC, IML, DEUSTO, TUE, IRIS) (M12-M30)

The aim of this part of the project is to assess and quantify the socio-economic impact of the previously identified scenarios of current and future trends and changes in international trade and SCs. To this end, in a first step, data on global trade patterns, such as Global Value Chain data, trade flows and the RWI's Container Throughput Index, are analyzed to provide a quantitative overview of recent developments and the current state of global trade patterns. To account for global value chains we have to disaggregate international trade flows. In addition, to get the appropriate level of disaggregation at the sectoral level we have to adjust the supply side of the country models. Moreover, it is necessary to modify the labor market module of the model to take into account the different productivities of households (taking inspiration from a recent work on minimum wage in Germany). With the help of this model, the effects of the changes in global SCs on sectoral value added in the EU countries and important trading partners can be analyzed. From this, effects on labor demand by skill level and the income distribution can also be derived. This will provide additional insights on the challenges that might result from changing global SCs for different regions and parts of the labor force. In particular the following steps will be done:

- Based on WP3 results, develop scenarios to quantify the impact of SC changes on socio-economic variables
- Collect data for international value chains and country level economic data to present descriptive statistics on therecent changes in SCs.
- Use the data about global value chains, sectoral economic variables and labor markets to calibrate our models. Wecollect data for EU countries and the important trading partners and competitors. The data is crucial for the choice of the level of aggregation of the model.
- Quantify the impact of SC changes on socio-economic variables of EU countries we construct a multi-country modelmulti-sector network model that include EU countries, trading partners and competitors.
- Implement different scenarios in the model and quantify the impact of SC changes on socio-economic variables.● Prepare and interpret the results to draw policy implications.

T3.3-Social accounting for sustainability (Task leader: DEUSTO; Participants: CNR, ZLC, IRIS) (M14-M30)

The main objectives of this task are the following: (1) to develop a model that calculate the value generated and distributed by a value chain in relation to the different stakeholders; (2) to develop a "self-assessment" tool that allows the companies of the value chain to identify its position regarding the social impact generated. This tool can also be used for carrying out prospective analyses about the effects of changes in the value chain; (3) to generate recommendations to public administrations regarding the actions that could be carried out to stimulate the value chain promoting a competitive sustainability.

To this end the research is focused in measuring the aggregated, traction and induced value of the value chain and how it impacts into the stakeholders, making possible to identify specifically the impact in areas such as gender, employment and well-being. The model and the self-assessment tool will be tested in four different value chains: The commercial Port of Algeciras, a Food value chain, a manufacturing value chain and a Public administration.

T3.4-The role of logistics outsourcing and impact on employment (Task leader: ZLC; Participants: CNR, IRIS) (M20-M30)

Environmental and social impact are now an important asset in business performance. This is especially true for logistics activities, where outsourcing is in place. Indeed, this task will tackle how logistics outsourcing has an impact on Corporate Social Responsibility, specifically in terms of employment conditions.

As an example, while the application of IT services in warehousing increases productivity, it highly depends on the degree of flexibility of the workforce. Task 3.4 will thus analyse the trade-off between productivity and flexibility and its impact on employment.
In order to do so, data will be collected from primary and secondary sources (i.e. interviews with trade unions and

managers of companies outsourcing logistics services; newspaper information and reports on the working conditions, etc.). A set of KPIs will be defined to measure impact on employment conditions, and recommendations will be derived.

T3.5- Digital Tool for social indicators visualisation (Task leader: IRIS; Participants: CNR, INESC TEC, ASTON, RWI, DEUSTO) (M18-M33)
The aim of this task is to offer a platform for information sharing and to improve the transparency of SCs with focus on the social dimension. It will serve as a "front end" and to integrate the analytical tools developed in previous tasks in WP3. In particular, a framework is designed and implemented to integrate and make available the most important results of the analytical tools developed and a web module will give access to selected results for different types of stakeholders. Specific profiles of access are created taking into consideration the information needs of different users, such as industrial companies, trade and social unions and policy makers. Information will also be made accessible to citizens to improve public awareness of how companies and countries are affected by the social impact of trade and for disseminating project results. The web module can be considered a prototype with a subset of the indicators to investigate the feasibility and usability of this kind of module. It is developed with an appropriate user friendly interface to help the user to navigate in an interactive and dynamic way through the selected results.

Work package WP4 – Development of policy scenarios

Work Package Number	WP4	**Lead Beneficiary**	2. INESC TEC
Work Package Name	Development of policy scenarios		
Start Month	18	**End Month**	36
Objectives			
WP 4 aims at proposing policy recommendations that can support public as well as private organisations not only in overcoming the economical, ecological, and social impacts from changes in European value chains, for instance because of the European Green Deal delivery, but also in realising the European Pillar of Social Rights Action Plan.			
Description			

Description of work

T4.1 Identification of key horizontal issues (Task leader: IML; Participants: Aston, INESC, TUE) (M18-M24)

- Transfer of the findings from WP 2 and WP 3 into a preliminary list of key horizontal issues
- Extension of the preliminary list through consultation with major stakeholders involved in the horizontal issues fromindustry, research institutions, associations, and government agencies.
- Extension of the preliminary list through review of roadmaps, research plans, research agendas, and reports fromfederations, national platforms, clusters, etc. referring to economical, ecological, and social impacts of SC changes ● Validation and clustering of the results together with various stakeholders

T4.2 Cross-country policy analysis (Task leader: INESC TEC; Participants: CNR, ASTON, IML, RWI, DEUSTO, TUE, IRIS) (M18-M26)

The identified KHIs from T4.1 are examined within EU and cross-country national policies, such as the European Green Deal and the European Pillar of Social Rights Action Plan. The goal of this task is to carry out a pan-European cross country analysis of the existing policies concerning SC, logistics policies, standards (e.g. Modern Slavery Act) in order to assess the regulatory readiness and the limitations of existing legislation. This is important because we need to explore the challenges and opportunities associated with existing legislation and policies. To perform this examination, the KHIs will be differentiated through the use of a weighted, static design structure matrix with the categorization of KHIs between clusters. In particular, the mapping will focus on aspects that can lead to a level playing field concerning European SCs and their synergies with industries, regarding economic, social and environmental concerns, requirements and standard practices.

An additional objective of this task is to design an analytics tool for assessing and monitoring the level playing field developments in key policy areas. This will be done by building a data driven model/simulation whose inputs will be macro-economic indicators (EU wide and country by country) for indicators linked for example to workers' rights (e.g. minimum wage, unemployment benefit, ..), competition and state aid (or subsidies for business), taxation, social, environmental protection. It will aggregate data from different sources (e.g. publicly available databases/information, as well as paywall/registration-wall information and EC online databases). The system can be developed using statistical and machine learning (AI) techniques in c++ and in Python notebooks (using libraries such as numpy, scikit, etc.). The combined approach will result in the identification of key factors, which, alongside the pairwise relationship between KHIs and policies, will serve as the foundation for the Policy scenarios building (T4.3) and the Policy design (T4.4).

T4.3 Policy scenarios building (Task leader: ASTON; Participants: CNR, INESC TEC) (M24-M30)

T4.3 uses the results from T4.2 to identify the areas where future policies will need to be developed in order to support the resiliency and effectiveness of post Covid 21st century SCs. Scenario development will be utilised to identify how policy will be shaped in the next years. Delphi method will be applied that delivers valid valuable, and reliable data for scenario building (e.g., Kameokaa et al., 2004, Rikkonen, 2005). Delphi method is multi-stage survey process (anonymous, written), where feedback of group opinion will be provided after each round to help us develop relevant policy scenarios. The outcome of this is going to be projections about how future policies can enable and secure

Objectives

Dissemination, Communication (D&C) and exploitation aims to implement a multitude of measures to communicate the findings of the project for an integrated approach of SC at European level as well as to exploit the ReSChaPe results. A sustainable dissemination and exploitation platform based on cluster involvement at a regional, national and European level aims to maximize the awareness of the project results and in consequence to facilitate the engagement of stakeholders in implementing them.

More specific objectives are:

- Create the platform that will enable to share knowledge and expertise gained in this project on how to generate andexploit synergy with industries, policy makers and social actors.
- Raise awareness and inform stakeholders about the project, the objectives, the results, and the relevance of the findings.The main effort will be devoted to disseminate the project results and communicate information about its objectives and scope across the European process industry community and EC stakeholders.
- Exploit project results implementing specific actions like dedicated training, consulting, joining other initiatives likeobservatories of SCs.

The dissemination and communication will be jointly developed and implemented with the results of the collaboration mechanisms with stakeholders and will support the collaboration mechanisms established there. The integration of the strategic seminar, the thematic workshop for each of future scenario at European level and the outreach actions on national and regional scale guarantee a maximum benefit for the project.

companies and its SCs with respect to critical social issues (e.g. the rights of workers and that supplier wages, benefits and working hours are fair and reasonable). More specifically we will develop some probable, extreme and unforeseen scenarios. One of the possible scenarios to be studied is related to EU self-reliance when there is a problem of resource provision (like energy crisis in Ukraine). For example, the energetic problem could be studied linking it to environmental sustainability, recourceefficiency of processes and implementation of circular economy in EU (as policies to put in place re-manufacturing of secondary raw materials, or supporting production of technologies for renewable energy), which will bring to the need to keep manufacturing here in EU and need new profiles, new type of labour forces, of workers.

T4.4 Policy design recommendations (Task leader: TUE; Participants: CNR, INESC TEC, ASTON, IML, IRIS) (M26M36)
For each of the key horizontal issues identified in task 4.1, the most policy-relevant economic, environmental, and social indicators will be identified. Each indicator will be mapped to a set of policies which have the potential to address the gaps in existing national and regional policies (Task 4.2). We will simulate the impact of policies on each indicator, and run sensitivity analyses for the most relevant variables. Simulation "what-if" scenarios will be performed using "Monte Carlo" method to find inputs for a given output (KPI) as well as forward-driven models (for a given combination of inputs, what resulting KPIs). Furthermore, this can be used as input to a game-theoretical approach to help identify winwin situations, zero-game, and so on. We will design consistent policy packages to address current gaps in economic and social policies. A workshop with a representation of all relevant stakeholders will be conducted to validate the potential impact of adopting such policies, and the perceived desirability of each resulting scenario from T4.3.

Work package WP5 – Dissemination and Communication

Work Package Number	WP5	**Lead Beneficiary**	4. ZLC
Work Package Name	Dissemination and Communication		
Start Month	1	**End Month**	36

Description

Description of work

T5.1 Dissemination, Communication and exploitation plan (Task leader: ZLC; Participants: All) (M01-M03)

The communication and dissemination strategy and plan will be developed with the aim of achieving the maximum involvement of partners, stakeholders, and will include:

- Definition of communication and dissemination objectives and corresponding target groups;
- Identification of the communication instruments (e.g. social media, websites) and dissemination tools (e.g.,conferences, magazines, online communities), including the specification of the dissemination and exploitation platform. ● Definition of dissemination methodology detailing the content and timing of the various dissemination activities, tasks and responsibilities assigned to the different involved
- Provision of structures for the communication and information actions/materials (structural design of the website,template, frequency and content of newsletters, standard lay#out for presentations and reports).

T5.2-Stakeholder engagement and networking with experts (Task leader: INESC TEC; Participants: All) (M03-M36)

Specific actions, such as workshops, webinars and participation in the project's community, will be performed towards business stakeholders (e.g. industry practitioners, policy makers, investors) to raise awareness on the scouted and validated solutions. The objective is to reach out to society in order to demonstrate the project's framework, results and its contribution to sustainable challenges (from the economic, environmental and societal perspectives). The communication with stakeholders focuses on the creation and development of joint business opportunities both regarding existing solutions and new policies. To this end, the project will integrate a 360º communication strategy that combines online and offline methods in order to have a broader reach both in terms of stakeholders and in terms of experts' elicitation activities. All stakeholders will be contacted through related partners and invited to become part of the project's community, acting as "supportive partners" for dissemination purposes in a win-win cooperation mode. This means that, at each communication milestone, the project will develop indications and materials required (e.g. e-mails, newsletters, social media messages, etc.) for the stakeholders to communicate with their ecosystems. In order to enhance stakeholder engagement, the project will offer: a communication platform for sharing their relevant activities with targeted audiences, visibility (including logo and website link) and exclusivity (participation in special activities for community members). Additionally, the project will offer multiple opportunities for engagement between stakeholders and experts in the form of dissemination and activity-related workshops. Their engagement is also of vital importance for the project's success, and will be achieved through constant invitation through related partners to the various fields of research and industry regarding the focus points of the project. These experts will be invited to participate in the project's community, and will have the opportunity to disseminate their knowledge with industrial practitioners, researchers, policy makers and other experts and stakeholders participating in the workshops and in the community.

In particular, we plan to organise a half-a-day cluster meeting and/or policy roundtable with your sister projects (RETHINK-GSC and TWIN SEEDS) and the EC policy officers in charge by M26.

The communication and exploitation platform developed in T5.1 will be used, which is going to serve as the main communication channel with stakeholders for dissemination of project results, new findings, workshop schedules and additional relevant information regarding the project development. The same platform is the foundation to bridge experts involved in the empirical activities of the project and the relevant stakeholders, therefore creating synergies that go beyond the scope of the project in order to foster increasing innovation and dissemination actions regarding resilient SCs.

T5.3 Dissemination, communication and exploitation actions (Task leader: ZLC; Participants: All) (M01-M36)

Dissemination actions related to promoting the project and its results will be implemented through strategic and targeted measures, possibly engaging in a two-way exchange with the stakeholders. Moreover, scientific communication of project results through participation into international conferences, workshops, seminars, and publication in scientific journals are planned.

Continuous dissemination of project achievements through the dissemination channels of sectoral communities and their related networks through newsletters, press releases and announcements of results in conferences and events will be managed. The task includes the search for opportunities to reach national and regional research policy makers to discuss the most relevant challenges of the coming years in a truly European context. A system of national/regional and sectoral relays will be set up to disseminate project results during the project and keep the network active after project end.

Communication actions will include a comprehensive fully public project website will be developed and linked to all partners' websites. It will serve as the reference point for people interested in the project itself, and will also contain reference material on all covered research fields, subscription functionalities for receipt of regular project publications and for participation in public project events, various contact options to the project management as well as selected experts.

Other communication tools will be: (i) the edition of a 6-monthly newsletter, which will include information concerning the project's activities and further related information such as relevant EU research policy and programme

developments, related events etc. in a concise and easy-to-understand way; (ii) supporting materials like logo, leaflets, brochure; (iii)

use of social media (LinkedIn, twitter, etc) as a way to involve stakeholders and to involve consumers; (iv) a 45-second promo video (outlining project objectives) by M6 and a 60-second promo video (summarising key research results and key policy recommendations) by M36.
In particular, it is planned to issue 2 EC policy briefs. Based on the results of WP4, it will be possible to provide recommendation on topics like trade, policy supporting new business models, supporting training, creation of ecosystems.

The task includes a final conference with wide participation from European industry, research community, policy makers and media will be organised at the end of the project in conjunction with the final General Assembly meeting of the project.

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (automatically posted online)

Sensitive — limited under the conditions of the Grant Agreement

EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Project and quality management Plan	WP1	1 - CNR	R — Document, report	SEN - Sensitive	3
D1.2	Data Management Plan and updates	WP1	4 - ZLC	DMP — Data Management Plan	SEN - Sensitive	36
D1.3	Progress report	WP1	1 - CNR	R — Document, report	SEN - Sensitive	36
D2.1	Report on trends, changes and challenges for SC of the future	WP2	3 - Fraunhofer	R — Document, report	PU - Public	8
D2.2	Report on SC strategies for resilience and sustainability	WP2	2 - INESC TEC	R — Document, report	PU - Public	12
D2.3	Report on Innovative ways to organize SCs	WP2	1 - CNR	R — Document, report	SEN - Sensitive	15
D2.4	Report on Identification of gaps in skills required	WP2	9 - ASTON U	R — Document, report	SEN - Sensitive	18
D3.1	Report on the impacts of COVID-19 on Supply Chains and social performance	WP3	9 - ASTON U	R — Document, report	SEN - Sensitive	15

D3.2	Report on model design for social impact and social accountability	WP3	5 - RWI	R — Document, report	SEN - Sensitive	18
D3.3	Report on results of the model Simulation	WP3	5 - RWI	R — Document, report	PU - Public	30

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.4	Report on develop model for social accountability and self-assessment tool	WP3	6 - UDEUSTO	R — Document, report	PU - Public	30
D3.5	Report on impact of logistics outsourcing on employment	WP3	4 - ZLC	R — Document, report	PU - Public	30
D3.6	Digital Tool for SC analysis with social indicators	WP3	8 - IRIS	OTHER	SEN - Sensitive	33
D4.1	Report on key horizontal issues and policy analysis	WP4	3 - Fraunhofer	R — Document, report	PU - Public	26
D4.2	Report on Policy scenarios development	WP4	9 - ASTON U	R — Document, report	SEN - Sensitive	30
D4.3	Report on Policy design	WP4	7 - TU/e	R — Document, report	PU - Public	36
D5.1	Communication and Dissemination Plan	WP5	4 - ZLC	R — Document, report	PU - Public	3

D5.2	Report on Dissemination, Communication activities	WP5	4 - ZLC	R — Document, report	PU - Public	36
D5.3	EC Policy Brief	WP5	4 - ZLC	R — Document, report	PU - Public	36

Deliverable – Project and quality management Plan

Deliverable Number	D1.1	**Lead Beneficiary**	1. CNR
Deliverable Name	Project and quality management Plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	3	**Work Package No**	WP1
Description			
This report will describe the measures to be carried out in T1.1. and in T1.2 in order to ensure an effective consortium coordination and project management. Quality levels and monitoring procedures are also defined.			

Deliverable – Data Management Plan and updates

Deliverable Number	D1.2	**Lead Beneficiary**	4. ZLC
Deliverable Name	Data Management Plan and updates		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1
Description			
DMP and its updated versions along the project in T1.3. Moreover, in this deliverable it will be reported the results of T1.4. due dates: M6, 12, 36			

Deliverable – Progress report

Deliverable Number	D1.3	**Lead Beneficiary**	1. CNR
Deliverable Name	Progress report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1
Description			
Report with main scientific deviations, generated EU-funded project synergies, remaining technical challenges and final policy expectations/scenarios.			

Deliverable – Report on trends, changes and challenges for SC of the future

Deliverable Number	D2.1	**Lead Beneficiary**	3. Fraunhofer
Deliverable Name	Report on trends, changes and challenges for SC of the future		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	8	**Work Package No**	WP2
Description			

This report will present the applied methodology as well as the identified trends, changes, and disruptions in T2.1 and the challenges to be faced by SCs identified in T2.2.

Deliverable – Report on SC strategies for resilience and sustainability

Deliverable Number	D2.2	**Lead Beneficiary**	2. INESC TEC
Deliverable Name	Report on SC strategies for resilience and sustainability		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP2
Description			
This report describes the results of T2.3 with a description of the SC strategies			

Deliverable – Report on Innovative ways to organize SCs

Deliverable Number	D2.3	**Lead Beneficiary**	1. CNR
Deliverable Name	Report on Innovative ways to organize SCs		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	15	**Work Package No**	WP2
Description			
Results of the T2.4 with details on the SC models			

Deliverable – Report on Identification of gaps in skills required

Deliverable Number	D2.4	**Lead Beneficiary**	9. ASTON U
Deliverable Name	Report on Identification of gaps in skills required		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP2
Description			
This report will present the skill set desired for SC professionals			

Deliverable – Report on the impacts of COVID-19 on Supply Chains and social performance

Deliverable Number	D3.1	**Lead Beneficiary**	9. ASTON U
Deliverable Name	Report on the impacts of COVID-19 on Supply Chains and social performance		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	15	**Work Package No**	WP3
Description			

This part will deliver a decision support framework that examines the linkage between SC risk management strategies and performance

Deliverable – Report on model design for social impact and social accountability

Deliverable Number	D3.2	**Lead Beneficiary**	5. RWI
Deliverable Name	Report on model design for social impact and social accountability		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
This report includes the preliminary result of T3.2 with design of Macroeconomic multi-county multi sector network model and of the T3.3 with designing preliminary data for the model of social accountability.			

Deliverable – Report on results of the model Simulation

Deliverable Number	D3.3	**Lead Beneficiary**	5. RWI
Deliverable Name	Report on results of the model Simulation		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP3
Description			
Final results of T3.2 after running and testing the model for the evaluation of the impact of new SC models on social indicators			

Deliverable – Report on develop model for social accountability and self-assessment tool

Deliverable Number	D3.4	**Lead Beneficiary**	6. UDEUSTO
Deliverable Name	Report on develop model for social accountability and self-assessment tool		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP3
Description			
Final results of T3.3 with the developed algorithm and the Self-assessment tool.			

Deliverable – Report on impact of logistics outsourcing on employment

Deliverable Number	D3.5	**Lead Beneficiary**	4. ZLC
Deliverable Name	Report on impact of logistics outsourcing on employment		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP3
Description			

Results of T3.4 with results of the model for evaluation of the impact of logistics outsourcing on employment of global SCs.

Deliverable – Digital Tool for SC analysis with social indicators

Deliverable Number	D3.6	**Lead Beneficiary**	8. IRIS
Deliverable Name	Digital Tool for SC analysis with social indicators		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	33	**Work Package No**	WP3
Description			
Report of the specifications, design, performance and validations of the digital tool for SC analysis.			

Deliverable – Report on key horizontal issues and policy analysis

Deliverable Number	D4.1	**Lead Beneficiary**	3. Fraunhofer
Deliverable Name	Report on key horizontal issues and policy analysis		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	26	**Work Package No**	WP4
Description			
This report will describe the identified and clustered Key Horizontal Issues for SCs from T4.1 and provide a review of EU cross-country policies from T4.2 considering challenges and opportunities for each KHI, as well as the mapping between KHIs and policies. This work will support activities in T4.3 and T4.4.			

Deliverable – Report on Policy scenarios development

Deliverable Number	D4.2	**Lead Beneficiary**	9. ASTON U
Deliverable Name	Report on Policy scenarios development		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP4
Description			
This report will provide the methodology and results for the identification of projections on future policies for sustainable and social SCs			

Deliverable – Report on Policy design

Deliverable Number	D4.3	**Lead Beneficiary**	7. TU/e
Deliverable Name	Report on Policy design		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP4

Description
This report is a policy brief containing a set of consistent policy measures to address current gaps

Deliverable – Communication and Dissemination Plan

Deliverable Number	D5.1	**Lead Beneficiary**	4. ZLC
Deliverable Name	Communication and Dissemination Plan		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	3	**Work Package No**	WP5
Description			
This report will describe the communication and dissemination strategy and plan for the project as from T5.1. In particular, it will include the plan for a 45-second promo video (outlining project objectives) by M6 and a 60-second promo video (summarising key research results and key policy recommendations) by M36.			

Deliverable – Report on Dissemination, Communication activities

Deliverable Number	D5.2	**Lead Beneficiary**	4. ZLC
Deliverable Name	Report on Dissemination, Communication activities		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP5
Description			
Final report of the D&C and exploitation activities of ReSChape results. In particular the results will include a 45-second promo video (outlining project objectives) by M6 and a 60-second promo video (summarising key research results and key policy recommendations) by M36.			

Deliverable – EC Policy Brief

Deliverable Number	D5.3	**Lead Beneficiary**	4. ZLC
Deliverable Name	EC Policy Brief		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP5
Description			
report with summary of the project findings to be delivered to EC policy services related to topics like supply chain strategies (also including EU self-reliance), models for supporting innovative eco-systems, how to enhance positive social impact. These EC policy briefs are expected to be issued at M15 and M36 (ZLC).			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Collaboration mechanisms established and dissemination plan defined	WP5	4-ZLC	Detailed plan of workshops and technical meetings defined. Creation of the expert group and commitment of most important stakeholders and dissemination plan defined (D5.1)	4
2	Trends and changes analysis	WP2	3-Fraunhofer	Main trends and changes influencing future SC scenarios have been identified and challenges in D2.1.	8
3	Innovative SC models	WP2	1-CNR	Innovative models are generated and are described in D2.3 and D2.4.	16
4	Advancement on models for social impact assessment	WP3	1-CNR	Verification on the advancement of the models expected in WP3, mainly design of the models, structuring dataset (D3.1, D3.2).	18
5	Check on implementation of D&C plan	WP5	4-ZLC	Expected dissemination activities have been realized, stakeholders, policy makers, external organisations have been reached by mean of the specific actions of the WP5 (D5.3).	18
6	Key horizontal issues for policy scenarios	WP4	2-INESC TEC	Identification of key horizontal issues and impact on policy actions have been defined in D4.1 and D4.2 as result of the related tasks.	24
7	Finalization of models for social impact assessment	WP3	5-RWI	Finalisation of the models expected in WP3. Testing and validation with stakeholders (D3.3, D3.4, D3.5). digital tools developed (D3.6).	33
8	Definition of policy scenarios	WP4	2-INESC TEC	Policy scenario generated and simulated (D4.2 and D4.3)	33

9	Dissemination plan realized	WP5	4-ZLC	Expected dissemination activities have been	36

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
				realized, stakeholders, policy makers, external organisations have been reached by mean of the specific actions of the WP5 (D5.2, D5.3).	

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	One or more partners do not comply with the work plan or is unable to provide the required contributions (LH = medium; S =high)	WP1	The Project Coordination team is committed to tightly monitor the progress of each partner and quickly react if required. If required, the consortium will consider transferring some activities to another partner or bringing a new partner into the consortium, upon consultation with the EU Project Officer.
2	Difficulties in being sufficiently innovative both from the methodological and the results point of view (LH=Low; S=high)	WP2, WP4, WP1, WP3, WP5	The methodology is well defined in the proposal and will be strengthen during the initial phases of each WP to have the full commitment of the partners. Partners will be monitored and corrective actions will be implemented if needed.
3	Difficulties in identifying trends and changes and extrapolating innovative SC strategies (LH=Medium; S=Medium)	WP2	Intensive brainstorming meetings and enlargement of experts to provide new ideas and to foster discussion. Intensive analysis of state-of-the-art reports concerning trends influencing SCs.
4	Delays in the design and development of the models, delays in transferring results from one WP to another, misunderstanding from one task to another (LH=Medium; S=Medium).	WP2, WP4, WP3	Project coordinator will monitor the involvement of all the partners according to the WP plan. WP leaders will have key role in promoting collaboration and monitoring their specific WP. Co-development will be supported by experienced partners from appropriate domains.

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
5	Difficulties to address and involve experts, other initiatives and public authorities, at European, national and regional levels. Stakeholder network does not represent all relevant stakeholders. (LH=Low; S=MEdium)	WP5	The project consortium is very well connected with a large part of the relevant stakeholder network. The completeness of the stakeholder network is checked periodically and adapted to the specific WP.
6	Stakeholders will not reveal information due to IPR issues. (LH=Low;S=Medium)	WP5	All considerations should regard data confidentiality will be taken into consideration at the beginning of the project both in the CA, in the DMP and in the privacy document to be signed by external participants.
7	Budget is not spent correctly or contributors do not deliver according to the plan (LH=Low; S=Low)	WP1	For these cases, the consortium Agreement will provide tools for the coordinator and the consortium to react in a timely manner and redistribute work and/or budget. Deviations from the work plan and budget will be identified on short term through the work in Task 5.2.
8	Delay propagation from technical WPs (LH=MEdium; S=MEdium)	WP1	Work is stringently and efficiently managed, planned and monitored. This will be done e.g. through a close follow-up by the coordinator and the review meetings with the WP leaders. The compliance has to be followed up through quality management.

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	Timing (month)	Location	Comments
RV1	18	Brussels	To be discussed with project officer
RV2	36	Brussels	To be discussed with project officer

Part B: technical description

***ReSChape* - RESHAPING SUPPLY CHAINS FOR POSITIVE SOCIAL IMPACT**

List of participants

Participant No. *	Participant organisation name	Acronym	Country
1 (Coordinator)	Consiglio Nazionale delle Ricerche	CNR	Italy
2	Instituto de Engenharia de Sistemas e Computadores, Tecnologia e Ciência	INESC TEC	Portugal
3	Fraunhofer Gesellschaft Zur Foerderung Der Angewandten Forschung E.V.	IML	Germany
4	Zaragoza Logistics Center	ZLC	Spain
5	RWI – Leibniz Institute for Economic Research	RWI	Germany
6	Universidad De La Iglesia De Deusto Entidad Religiosa	Deusto	Spain
7	Technical University of Eindhoven	TUE	The Netherlands
8	Iris Technology Solutions	IRIS	Spain
	Aston University (associated partner)	Aston	UK

N.B. based on the letter received from REA-Ref. Ares(2022)2815306 - 08/04/2022, the applicant established in the UK, namely Aston university, will participate as an associated partner, in agreement with the rest of the consortium. Aston University will fund its tasks without EU funding and the total EU funding to the project will be reduced accordingly. According to the communication from REA, Aston University may be later re-established as beneficiary through a grant amendment requested by the consortium if (i) the instrument of association with UK becomes applicable, (ii) this association agreement provides for its retroactive application prior to the start of the action, (iii) there is a corresponding financial contribution from the UK to the relevant call. We accept that subject to the fulfilment of these conditions, the granting authority may approve reimbursement of the costs incurred from the

beginning of the action. In its role of associate partners, Aston engages to undertake the activities described and agreed in this DOA, and their participation will be ruled in the Consortium Agreement.

History of changes at 29.04.2022

Pag.	Modifications
List of participants	Shift of Aston to last line, as associated partner
Ch.1.2.1	Added the possibility to study other scenarios related to supply chain autonomy / selfreliance as a way to reduce/limit the linksand dependences from other global actors.
Ch.2.2.	Revision of the chapter to give evidence to dissemination and communication actions and add related KPIs
Fig.3.1	Modification of fig.3.1 to change the Gantt. Anticipation start date of T3.5 (M18 instead of M22) to give more time for the development and anticipation of end of T4.2 (M26 instead of M28) to release D4.1 to T4.3 and T4.4.
WP1 description	Added T1.4 for Ethics compliance
WPs description	To reduce number of deliverables we have decided to merge the following: • D1.1 and D1.2 integrating project management plan and quality control plan at the same time. • D2.1 and D2.2 because the topics are strictly correlated and tasks finishing same time. • D3.2 and D3.3 because they are related to preliminary results of 2 tools on social impact and they are due at the same time. • Shift delivering of D3.6 to M33, since this is the closing date of the related task. • D4.1 and D4.2 because topics are complementary each other and preparatory to T4.3 and 4.4.
WP5 description	Added 2 policy briefs in WP5 dissemination, at M15 and M36 as from PO request with some examples of possible topics to be included.

Table 3.1j	Change table 3.1j with new distribution of in-kind contributions for the following reasons. Updating the table 3.1j, INESC TEC decided to add IPP's (Instituto Politénico do Porto) contributions in kind. The entities UP and IPP will not be involved in the project, and no activities will be carried out by them. INESC TEC envisages the use of contributions in kind provided by UP and IPP regarding secondment of persons free of charge (Article 9.2 of the General Model Grant Agreement). The inclusion of the IPP is due to the addition to the project team of three Professors/Researchers of this institute who have recently collaborated with INESC TEC: Hélio Cristiano Castro, Nelson Jorge Ribeiro Duarte and Cristóvão Dinis Polido Sousa. They have been developing research on the relationship between the adoption of digital technologies and sustainability in the context of supply chains and will contribute to activities related to WP2 and WP4. Professor Maria do Rosário Moreira from the Faculty of Economics of the University of Porto has researched in areas such as supply chain management, logistics and quality management, and has experience in sustainability issues in supply chains. She will mainly contribute in WP4. Although not contemplated in the proposal phase, INESC TEC believes that the inclusion of IPP members in the team will be beneficial due to their experience in project-related topics, and because they have the potential and availability to participate in different phases throughout the entire project. On the other hand, the participation of University of Porto will be reduced compared to the original plan in the proposal, as it will be focused on WP4. Finally, as explained in Table 3.1j, INESC TEC reinforces that "the researchers are authorized to do their research work for INESC TEC and using its research resources, such as offices and laboratories, including a clean room, meeting rooms, communication network
	and other supporting infrastructures. For all contractual purposes, INESC TEC assumes full responsibility for the involvement of these researchers in the project, since they are members of its research team.
Ch.3.2.	Consortium as a whole: shift Aston to last organization in the list as associated partner.
Annex I	Added annex I explaining collaboration with affiliated partner for CNR. Under the framework agreement existing between CNR and Università di Padova, it is planned to have a collaboration for the project.
Annex II	Added annex II with budget for Aston

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1. Excellence

1.1 Objectives and ambition

1.1.1 Motivation

The last year and a half was characterized by a series of sudden changes that heavily affected the economy and the whole of society worldwide and that will have serious consequences in the coming years. While the shock and emergency of dealing with the immediate health crisis have taken the windward, the entire world had to work on the necessary containment measures to save lives and avoid any further overload that health system had with a devastating impact on the global economy.

Moreover, considering to what extent economy and wealth of each state are based on international trade and how global the scale of production chains has become, such a large freeze of trade in some goods typologies (such as clothes, furniture, cars, etc.) impacted with double-digit loss ratios on the real economy. For Europe, the statistics report 16 billion euro loss, of which 2.5 in the automotive sector, 5.2 in the mechanical sector and 2.6 in the chemical sector with a need to completely rethink the supply paths.

According to the OECD, the COVID-19 epidemic will curb the global GDP growth by half a point, halting at 2.4%, while the UNCTAD estimates at USD 1 trillion the cost of the global economic slowdown. In the first quarter of the year, GDP dropped 5.3%, marking a decrease in value added of 1.9%, 8.1% and 4.4% in the agricultural, industrial and services sectors respectively. At European level, GDP is estimated to drop by 7.7%, while in the USA it dropped 6.5% (ISTAT, 2020). A study carried out by the European Association of Automobile Manufacturers (ACEA) found that the European workers in the automotive sector affected by production stops were about one million, out of a total of about 2.6 (Motore, 2020). Car sales in EU is expected to drop by more than 3 million units, from 12.8 in 2019 to around 9.6 million units this year (ACEA, 2020). At global level, the production of light vehicles is expected to drop by about 11 million units (from 88.9 million in 2019 to 77.9 million in the current year) (Deloitte, 2020).

As Europe enters a new decade, progress still needs to be made to reach high levels of employment, skills and employability, and robust social protection systems. In December 2020, 16 million people were out of work and youth unemployment was at 17.8%, considerably higher than general unemployment. Low-skilled, low-paid workers, and temporary workers were the first to be laid-off due to the COVID-19 outbreak. Industrial companies, policy makers and social actors need to put together their strengths to define a new collaborative model where the competitiveness is based not only on technological and economic advancement but also on social cohesion and environmental sustainability.

1.1.2 Objectives

The recent pandemic has completely changed the globalization paradigm expanding the borders of value and supply chains (SC) with a shift in trade patterns and consumer dynamics. On the one hand, the acceleration of digital technologies and the reduction of obstacles to international trade facilitate the integration of EU companies in global value chains, on the other hand, the profound transformation of global value chains has raised significant social, economic and environmental challenges, including increasing divergence in productivity, labour market effects, working conditions in the progress towards resource-efficiency and decarbonisation, with problems of security of and access to strategic supplies.

For this reasons, the aim of the ReSChape project is to propose new strategies and policies to reshape SCs assuring a positive impact on social indicators. Both macro and micro-dimension of the problem will be considered integrating approaches from different disciplines. In particular, the objectives of the proposal are:

- *Objective 1*: To analyse and identify social, economic and environmental changes and disruptions (including coronavirus) and evaluate their impact on SCs, identifying related challenges in terms of relationship between countries (taking into consideration level playing field issues), configuration of the network, and impact on employment.
- *Objective 2*: To study and propose a set of SC models for the evolution of global SC integrating strategies like resource efficient, closed-loop and humanitarian as a way to increase EU resilience and sustainability. Particular attention will be given to the role of digitalization as a way to establish new paths for social inclusion taking into consideration the needs of urban and rural areas. Some important European sectors like fashion, automotive, medical and machine tools will be analysed with case studies and survey.
- *Objective 3:* To develop Innovative tools for monitoring and assessing sectoral trade patterns and defining mechanisms to evaluate relationship of disruptions like pandemic and global value chains taking into consideration impact on employment, economic growth, incomes etc also in the long term. Moreover, it will be analysed the impact of different trade patterns, on the EU value added of sectoral and countries with a specific focus on analyzing income inequalities and proposition of decent work and social cohesion taking into consideration also gender issues and social disparities.
- *Objective 4*: To develop innovative policy scenarios with recommendations for future global value chains: policy scenarios will be based on key horizontal issues impacting on several sectors and will provide recommendations for EU, national and sectoral strategies, policy measures and targeted actions aimed at shaping fair, inclusive and sustainable trade patterns, value and SCs as well as production networks.

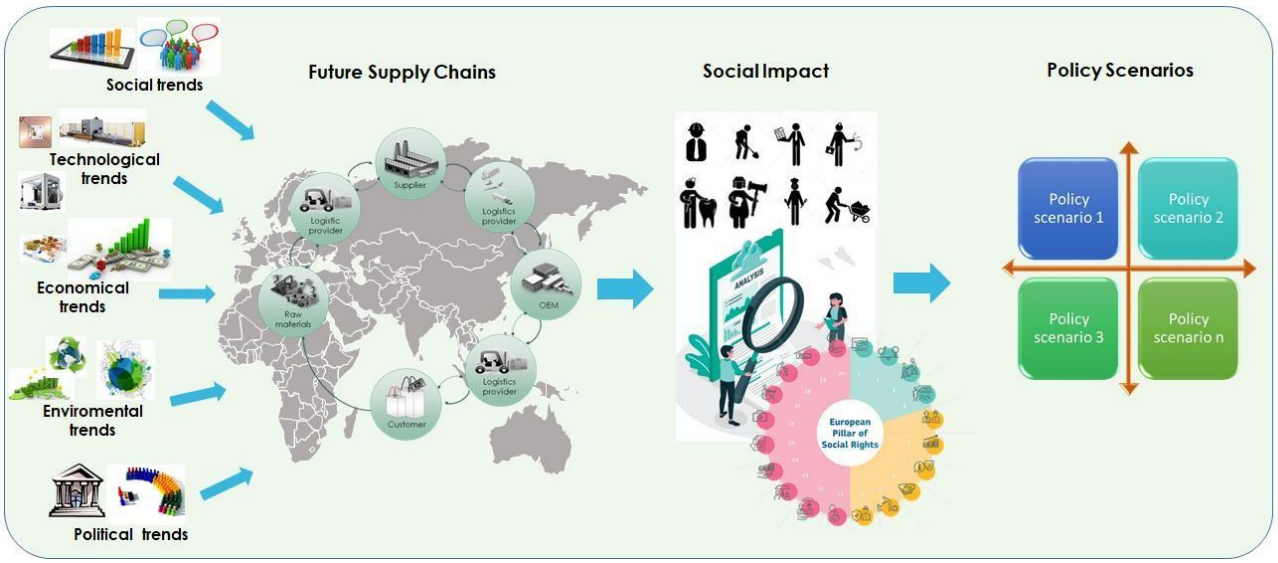

Figure 1.1 ReSChape Project overall concept

The table below represents the ambition of the project objectives and some key performance indicators (KPIs) are reported as measures of success for the short term, after the end of the project activities. Also relevance to the work programme is reported while long term indicators are reported in the section 2 of impact.

Table 1.1 Objectives and ambition

Objective	Ambition & KPIs	Relevance to the work programme
Obj1. Trends , changes and disruptions and their impact on SCs	ReSChape project analysed more than 70 global trends, isolates those with an impact on social and sustainability and derives the related challenges for SCs.	The identification of trends and their impact on global SCs forms the essential basis for developing and analysing sustainable SC models and making recommendations to ensure resilience and sustainability.
Obj2. Innovative SC models for social resilience and sustainability	ReSChape project engaged directly 100-150 industrial companies, active in the European SCs in 6-7 relevant sectors, and defined 4-5 SC models.	"Research should .. conceptualise the actual global and sectoral trade patterns, value chains, SCs and production networks in light of the EU's long-term policy priorities of social resilience and competitive sustainability"
Obj3. Assessment of impact of trade patterns and new supply chains on social indicators	Develop 4 models (micro and macro level) to analyse social impact. Modeling up to 40 countries an 30 sectors per country. Involve more than 100 experts. Analysed the 20 EU Pillars on Social rights (EU2021) and identified 10-20 social indicators to be structured for tracking them with the project models.	"then analyse the impacts of different trade patterns, value chains and production networks on the EU value added, labour market, income inequalities, decent work and social cohesion in urban and rural areas, taking into account gender differences."
Obj4. Policy	Mapped and identified 6-7 Key	"develop a comparative assessment with the main
scenarios for positive social impact including level playing field developments.	Horizontal Issues related for example to trade conditions, taxation, international agreements, and defined 3-4 policy scenarios. Simulation and what if analysis should bring to an index of consistency of the retained scenarios for over 90%. Involvement of more than 40 social actors, and more than 40 policy makers.	strategic partners and provide innovative, forwardlooking policy scenarios with recommendations for future global value chains, trade patterns and trade intensities. The policy scenarios should take into consideration analytical approaches, which will improve the economic and environmental performance of SCs in the EU. The policy scenarios and recommendations should focus on EU, national and sectoral strategies, policy measures and targeted actions aimed at shaping fair, inclusive and sustainable trade patterns, value and SCs as well as production networks."

1.1.3 Progress beyond the state-of-the-art

While the Fourth Industrial Revolution has a high potential to trigger social changes of a variety of natures, the use of digital technologies can be used to address social challenges. The concept of Industry 5.0 is not just a chronological continuation of existing Industry 4.0 but is conceived to highlight the importance of research and innovation to support industry in its long-term service to humanity within planetary boundaries. Technologies can be used in the context of social innovation and the circular economy in a variety of ways and their contribution has been highlighted in recent years by managers and researchers. Fig. 1.2 shows a set of technologies identified as relevant for Next Generation Supply Chains but new research is necessary to study how they can contribute to a positive impact on social dimension toward circularity of industrial value chains (Fornasiero et al., 2021).

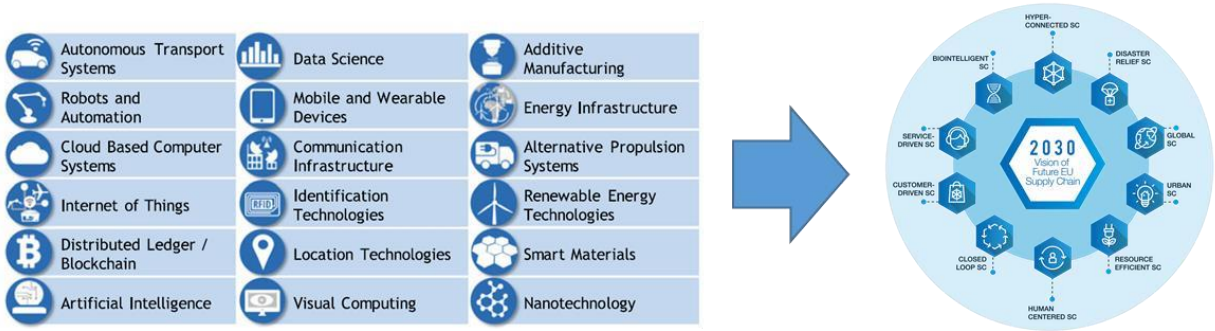

Figure 1.2 Technologies for Next Generation Supply Chains (Fornasiero et al. 2021)

In particular we can consider the project has the ambition to propose novel concepts and approaches in the following areas:

Supply chain models for social resilient and sustainability

Nowadays, Supply Chains (SC) have to be more and more able to adapt and identify possible future evolutions to preserve their competitiveness and meet the challenges derived from the complex scenario. (Ivanov and Dolgui, 2020; Cannella et al., 2016). Some important drivers of change affecting the global scenario are related to the social dimension, like changes in demographic, migration flows, new customer habits and also to the environmental dimensions like climate changes and depletion of natural resources. The transformation driven by the growing adoption of Industry 4.0 technologies represents at the same time a challenge, due to difficulties in its adoption, especially for what concerns requested skills, trust and security, and an opportunity to be leveraged by industry and related SCs (Ivanov et al., 2019) to enable social changes both in business and in society. In the last two years, the pandemic caused by Covid-19 and related severe measures to contain the contagion have forced the shutdown of many sectors of the economy, with severe interruptions into global SCs. In addition to these drivers, European society is also pushed, in line with initiatives such as the EU Strategic Agenda 2019-2024 and the European Pillar of Social Rights, for a paradigm shift towards the green and digital transitions, leaving no one behind. In this context, Europe needs to further strengthen its resilience, in order to be able to face and react to future shocks and assure citizens higher social standards and efficient and safe employment.

From pure automation and connectivity, the evolution towards Industry 5.0 (EU2021) envisaged in last year by EU will have to embrace a synergy between humans and autonomous machines (Nahavandi, 2019, Nair et al., 2021) with the possibility to make the difference in approaching social dimensions like employment, decreasing disparities, increasing safety. In order to ensure that both companies and people benefit from the digital transition, rethinking and redesigning SC is necessary and workers as well as citizens in general should be involved in all the steps of the digital transition. Special attention needs to be paid also to the to the creation and consolidation of SCs for the enhancement of territorial development in rural areas, assuring the sustainable management of natural resources and the creation and maintenance of employment, according to The EU's Rural Development Policy 2014–2020 (Scaramuzzi et al. 2020).

ReSChape project will study innovative SC models for an advanced pace of technology adoption, where humans will play active roles (Reiman et al. 2021). It will study how SC configuration and related processes need to change to assure that employees are at the center of the business thanks to new digital solutions for ergonomic systems, knowledge management, new human-machine interaction to maximize effortless and safe work, where inequalities are left behind thanks also to new training tools (Siemieniuch et al., 2015). The main actors of the new models are indeed citizens that will play a leading role, for what concerns the transformation of urban and rural areas, and with the collaborative processes enabled by social media they can act as innovators to create the desired environments where to balance working and life time. (Deguchi, A., & Sameshima, 2020). In light of the envisaged scenario, the project aims at identifying a set of SC models as prominent to face the variegate pool of drivers of change challenging global SCs and to support the transition towards a digital and sustainable society.

Methodologies to assess social impact of trade and of new supply chain models

Assessment of the social impact will be based on the application of different methodologies in order to analyse relationship between several different dimensions both at macro and micro-level taking into consideration point of view of SCs and of policy makers as well. In particular:

- **Assess impact of COVID on supply chains and social indicators**

Regarding the impact of Covid-19 and globalisation, on labour markets in terms of social performance indicators (i.e. unfair wages, child/forced labour, discrimination, excessive working time), in the last decade, a lot of studies have been conducting on sustainability in SC management however, socially related aspects are still neglected such as labour exploitation. With the advent of the novel coronavirus pandemic, the use of child labour in agriculture, mining and other sectors raised and led more children to work as a result of school closures, economic insecurity and

restrictions on trade and movement. However, it has been argued that increased openness and foreign direct investment due to globalisation can lead to keep labour costs low and at the same time to have high % of child labour (Palley, 2002). Several companies involved in sub-contracting of development, manufacturing of footwear, apparel have involved in various scandals in their SCs e.g. discrimination, excessive working time.

Thus, in this project we will fill the gap of identifying the impacts of these megatrends on social performance indicators. In particular, the impacts of job satisfaction, commitment (believing and accepting its goals and values and investment of effort on its activities), Organizational justice (if the employees have been treated fairly in their jobs), CSR attitudes, on SC social performance (i.e. unfair wages, child/forced labour, discrimination, excessive working time).

- **Evaluation of social impact of trade and new supply chain models**

COVID crisis has once again highlighted the importance of international value chains for national economic development (OECD 2020). Even before that, the development of global value chains was changing the economies of Europe (ECB 2019). A number of studies have already examined the importance of SCs for the transmission of disruptions. Acemoglu et al. (2012) use an Input-Output model to analyze the propagation of sectoral shocks. They find for the US economy that sectoral shocks are able to produce sizable fluctuations in overall economic activity. Using a similar model Acemoglu and Tahbaz-Salehi (2020) show that firm failures and resulting disruptions of SCs are also able to amplify shocks. This highlights that changes in SCs can have economic effects that are visible at the national level. Huo et al. (2021) extend this network-model to the international level. These authors construct a multicountry, multi-sector network propagation model. For all countries each sector is represented by a production function that uses capital and labor as well as intermediaries. Intermediaries are a bundle of inputs potentially from all sectors and countries. In this model homogeneous households supply labor to each sector separately. The model of Huo et al. (2021) will be used to analyze the quantitative effects of SC changes on national economies. However, to analyze the questions of interest in this project some modifications are necessary.

In this project, we are focusing more on EU countries, important trading partners and competitors with regard to the location of production sites. To keep the model manageable we will choose a level of aggregation that is simple but sufficient to answer the relevant questions of this project. To analyze the economic effects of changes of socioeconomic variables we have to modify the model structure with regard to the household sector and the labor market. In particular, we have to distinguish different households that offer different qualities of labor. This allows us to analyze the effects of changes in SCs on job qualities and income distribution. There is some evidence, that there is an increasing skills mismatch, in particular skills shortages and overqualification (Vandeplas and Thum-Thysen 2019). It can be assumed that this is to some extent related to changes in SCs affect the location of jobs (Kizu et al. 2019).

- **Social accounting for sustainability**

Three theoretical frameworks are distinguished in the social impact assessment knowledge area: (1) The Social Balance, sometimes known as social accounting, with a great development in France and Spain in the 90s of the last century. (2) The impact analysis whose main instrument has been the Social Return on Investment (SROI), developed for public investments in Great Britain and later generalized in the US in the specific field of non-profit organizations. (3) The triple Bottom line and the instruments derived from it, the Global Reporting Initiative (GRI) and the Integrated Reports being the most significant. All of them suffer from important limitations. For example, the social balance sheet, the GRI and the integrated reports, work with a wide battery of indicators with the impossibility of reducing them to a single unit of measurement and, therefore, the impossibility of integrating the analysis; Also, the SROI, which, similar to the ROI, only works well in the analysis of investments, that is, of funds released with alternative investment options, producing important imbalances in its application to current spending (Retolaza and San-Jose, 2018).

In this context, the social accounting model is developed, to which the term "Monetary" is sometimes added, precisely to differentiate it from the Social Balance, which is considered non-monetary social accounting. Its theoretical origins at the first Centre for Social and Environmental Accounting Research (CSEAR) (Gray, 2001), consolidating methodological proposal through practical application in multiple companies, which have been happening since 2010

(Retolaza, San-Jose and Ruiz-Roqueñi, 2016). At this time, the Spanish Association of Accounting and Business Administration (AECA) is integrating Social Monetary Accounting with the Comprehensive Report and proposing its use to the EU, through the European Financial Reporting Advisory Group (EFRAG), as a reference for the mandatory sustainability report (previously Non-Financial Information). At this moment, the application of social accounting to different types of value is relatively well solved (Retolaza and San-José, 2021): added value, induced value, non-market value, and even emotional value; but it is only possible to apply it to the first level of the value chain of the analyzed organization.

ReSChape will face the challenge to enlarge the method to the entire value chain. That is, to include all the organisations involved in the SC, identifying its generated social impact into different areas such as gender, urban and rural areas, well-being, employment, environment and territory development. Both macro- and microeconomic approaches will be considered in developing an analytic and innovative approach for social accountability applied to global value chains.

- **The role of logistics outsourcing and impact on employment**

RECOVER project highlighted how, in recent years, there has been a shift in the logistics sector towards higher added-value services, as a consequence of automation and digitalization. This ultimately translates into employees for warehousing and distribution being outsourced. This new situation in the sector impacts collective bargaining coverage. Indeed, whereas the majority of in-house logistics manpower is covered by sectoral and company collective agreements in manufacturing, wholesale and retail, only a small share of outsourced workers is covered by them. Even though there have been attempts by social partners to organize logistical services as a new sector to negotiate, it is difficult to characterize it as a sector as it is a horizontal activity across sectors.

On top of the above, trade unions are also faced with different challenges around organizing workers, such as a high share of temporary workers, periodic expiration of contracts between suppliers and contractors, and fragmentation of workers along a service chain. Additionally, as logistics is a low margin sector, if working conditions are improved in one company along the SC, the contract risks not being renewed in favour of a cheaper competitor.

ReSChape will face these challenges by proposing cross-sectoral cooperation along the SC, as well as targeting newly developing sectors, such as logistics as a service and e-commerce.

Policy scenario building

Policy scenario analysis comprises a set of methods which aim to predict the impact of policies on a desirable metric, such as energy consumption or factor input productivity. By creating functional relationships between policies and key parameters in scientific models, scenario analysis does not provide a unique prediction about the future, but rather offers a set of possibilities about how the world may evolve over a certain time frame (Tourki et al., 2013). In recent years, the use of scenario analysis has become widespread in sustainability studies. Scholars have used scenario analysis to investigate issues such as waste management (Andreoni et al., 2015), the efficacy of feed-in-tariffs in the energy sector (Kishita et al., 2017), or vehicle pollution control measures. Furthermore, the European Commission has actively been using scenario analysis to analyze how to successfully deliver the goals agreed in the European Green Deal.

To become a sustainable society, governments need to pay attention to the three dimensions of the so-called triple bottom line: people, planet, and profit. However, the vast majority of studies using scenario analysis focus only on one of the three dimensions, often the environmental one. This simplified view may not represent the true interests of all stakeholders. In fact, in some cases, issues related to social sustainability might be a more important decision driver than economic or environmental factors (Adaloudis et al., 2021). Hence, to become more realistic, it is critical that scenarios incorporate all three dimensions. *For this reason, the project will also produce a set of policy recommendations and policy briefs to design consistent policy packages to tackle economic and social gaps within policies.*

1.1.4 Positioning in terms of research and innovation maturity

According to its objectives, ReSChape R&I maturity is mainly related to develop studies and propose models which shall change the theory on SC, on Social Impact assessing and Monitoring, as well as on Policy Scenario Building as well as propose methodologies, best practices, recommendations to be applied by industrial companies, social actors,

policy makers to improve the understanding of how trade flows of the new normal can impact on the social dimensions. Therefore, R&I maturity is situated in the spectrum from 'idea to application' as it starts from existing models both developed by partners and found in literature, and will match with identification of industrial approaches (idea) to propose innovative ways to deal with these kind of challenges aiming to their application.

1.2 Methodology

1.2.1 Methodology applied in ReSChape project

This section explains the methodology the consortium will use in ReSChape project. The project activities are structured to integrate different approaches both from organizational studies, management theory, econometrics, policy economics, as well as simulation with analytics and what-if analysis applied to policy scenario building and social impact evaluation. Fig.1.3 shows how the ReSChape objectives (light blue boxes) are developed in the project and intertwined each other. Methods and tools are also shortly summarized (green boxes). In the following paragraphs, each of the boxes is explained.

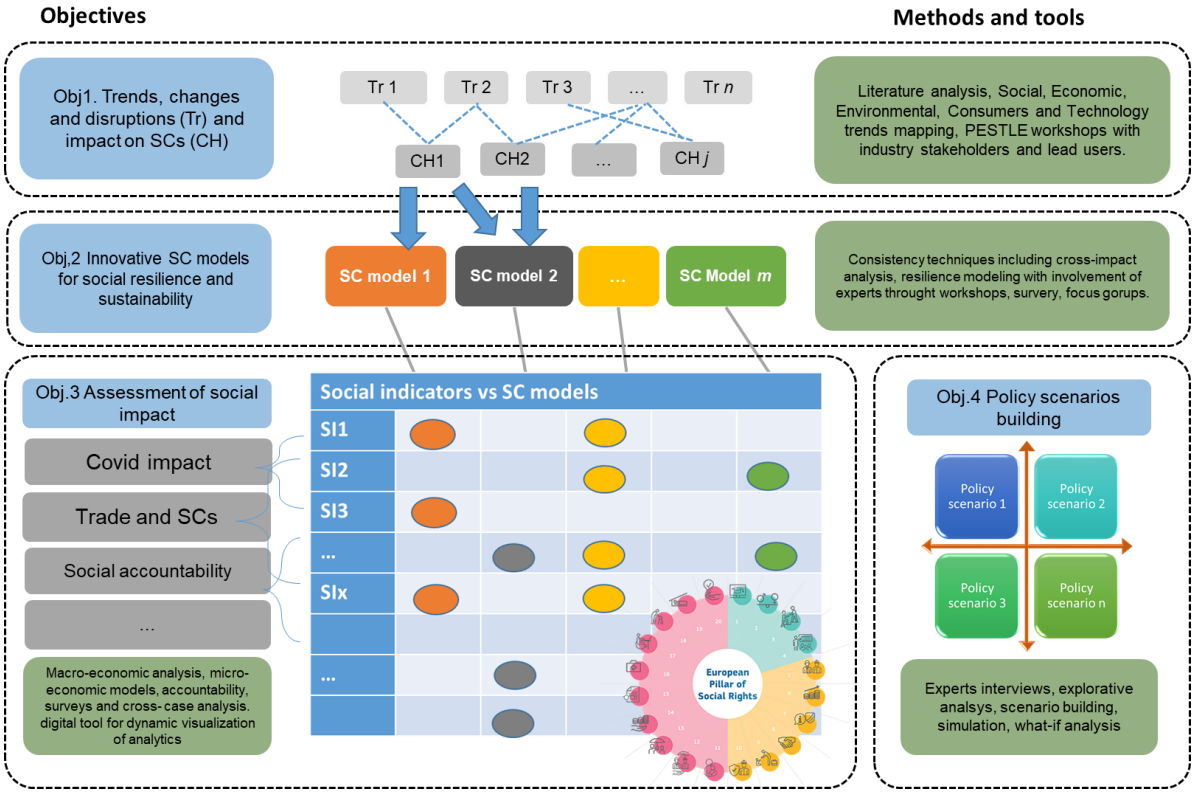

Figure 1.3 Overall methodology of ReSChape project

Obj.1 Trends, changes and disruptions and their impact on Supply chains

The definition of innovative models for social resilient and sustainable SCs will be based on an iterative process, since it is required back and forward interactions between research team and experts in order to achieve consistent results. This part of the work will start with the identification of trends, changes, and disruptions that have economical, ecological, and/or social impacts on European SCs, e.g. scarcity of natural resources, natural catastrophes like Covid-19, and demographic change. A PESTLE analysis will be performed to investigate trends, changes, and disruptions originating from the following dimensions: Political, Economic, Social, Technological, Legal, and Environmental. The basis for this analysis is provided by the results of the NEXT-NET project, in which a total of 23 megatrends and 72 trends impacting future SCs were identified and will be here updated with a particular focus on their influence on social resilience and sustainability.

Moreover, the current insights into the state of affairs, including but not limited to Covid-19 and the respective changes, e. g. to the design and operations of global SCs as well as consumer behaviour, will be included. This supplement will be based on a systematic literature review, for instance according to Denyer and Tranfield, and a workshop with experts from industry, associations, and policy makers. In this way, findings grounded on the contributions from several stakeholders can be integrated and thus enable coherent conclusions on how trends, changes, and disruptions impact and create new challenges to the social resilience and sustainability of European SCs. The interaction with experts will be organised to be open-minded, with an undirected atmosphere to facilitate the generation of fast-result insights (Zeng et al. 2019). Partners will also map digital technologies from literature and projects to study how they can trigger a new social resilience through the proposal of novel SC models.

Thematic analysis will be used as a qualitative research method for identifying, analyzing, organizing, describing, and reporting themes found within the collected data (Braun et al. 2006). This methodology is also useful for handling and summarizing key features of a large data set, helping to produce a clear and organised output (Nowell et al. 2017) through: acquaintance with data (identification of challenges through careful reading of the data to familiarise with the content); generating initial codes to simplify and focus on specific characteristics of the data to symbolically assign a summative and essence-capturing attribute to data (Saldaña 2009); sorting and collating all the codes to extract the summary issues with an inductive approach; proposing of SC strategies.

Obj.2 Supply chain models for social resilience and sustainability

Development of innovative ways to face social challenges will be based on theories like: Resource-based view (Barney, 1991), to explain how strategically managed interactions contribute to gaining competitive advantage; dynamic capabilities (Teece et al., 1997), and contingency theory (Donaldson, 2001) to study how companies can adapt, integrate and reconfigure their organizational skills, resources and competences to match with changing environments and to be able to obtain a sustainable competitive advantage; panarchy theory (Gunderson & Holling, 2002), rooted in ecology, to provide a structure for understanding how a system follows the movements of adaptive cycles on scales of time, space, and meaning. Basing on these theories some of the steps to be done to develop innovative SC models will work on the following concepts:

- Social resilience till now has been studied mainly as the adaptive and learning capacity of individuals, groups and institutions, or the entire society, to self-organise in a way that maintains system function in the face of change or in response to a disturbance (Mclean et al, 2013, Obrist et al., 2010). On the other side, social Supply Chain models are mainly focussed on consumer involvement (Fletcher et al., 2016), social sustainability practices, as health and well-being of people in the supply chain and impacts on society (Marshall et al., 2014) but no link has been yet clearly created with the industrial world and the role business can have to help social resilience. Moreover, little has been investigated on the social risks in which the supply chain may be involved (i.e. worker conditions, child labor by large brands) (Cunha et al., 2019). For this reason, Reshape solutions will help to pave the way towards novel SC models to support social resilience focusing on specific actions and practices at the 3 levels of intervention (readiness, response and recovery) (Han et al., 2020; Krammer, 2021).

- Proposing innovative SC models. As an example, models like resource-efficient and biointelligent SCs are promising to enable not only (self-)adaptive, but also environmental and social sustainable SCs in Europe. At its core, this approach uses insights from nature and integrates them into products and processes, such as swarm intelligence, circular economy, and bionics. By doing so, not only resources are conserved but also rapid responses to disruptions in global SCs, e.g. caused by Covid-19, and corresponding changes in trade and consumer behaviour are enabled. Therefore, ReSChape will studySC models like biointelligent from the point of view of the major impact on European SC performance in terms of ecology and society. Moreover, it will be studied how the identified SC models can contribute to preservation of European competitiveness and the associated safeguarding as well as creation of workplaces, and how they can contribute significantly to the realisation of the European Pillar of Social Rights Action Plan. Other SC models will be studied considering feasible geopolitical scenarios in which the EU aims for supply chain autonomy / self-reliance as a way to reduce/limit the links to Russia, China and maybe other global actors.

- Studying how the shift toward automation could cause a misalignment and shortage of skills needed by companies to operate in a more automated, digitally enabled environment. Indeed only 44% of the companies declared that their employees were prepared for digital innovation in the SC (EY, 2021). In this study, we will explore the changes in skills requirements in the logistics and SCM field to deepen the understanding of the sources and mechanisms of wage inequality." In particular, the project will investigate discrimination and skill differences to solve these disparities. For example, in England/Northern Ireland and the US, the wages of highskilled workers are twice as high as those of low-skilled workers (Broecke, 2016) highlighting the different education background can be attributed 83% of the wage gap between workers with different levels of parental education and 72% of the wage gap between native- and foreign-born workers. On a similar note, wage gap derived from skills differences account 23% between men and women. Several companies realised that in order to create a more equitable and inclusive society, they need to prepare people for the future of work due to expected related changes that dramatically impact the employment prospects. For example, Unilever will equip 10 million young people with the skills required to protect their livelihoods and prepare them for better job opportunities by 2030.

- Implementing the principles of Industry 5.0 to help the transition and supporting industry in its long-term service to humanity and society. Contributions to Industry 5.0 will address increased need of collaboration between humans and smart systems allowing to merge the high-speed accuracy of industrial automation with the cognitive, critical thinking skills of humans (Atwell, 2017, EU 2021). Beyond manufacturing boundaries and according to the concept of Society5.0, this includes the vision of a new human-centered society, where advanced technologies are applied in everyday life, and in different spheres of activity, to provide products and services satisfying

various potential needs as well as reducing economic and social gaps, for the benefit and convenience of all citizens. (Fukuda, 2019, EU2021). The project will propose a set of SCs to enhance the active role of stakeholders such as workers, consumers, citizens and leveraging on digital technologies to enable manufacturing industry contribution to this essential change of pace (Fornasiero and Zangiacomi, 2021).

Obj3. Assessment of Social impact of trade and new supply chain models

ReSChape proposes a set of modules to assess and monitor social impact of changes in trade and in SC models that range from macro-economic analysis where country and sectoral vision is provided to specific SC analysis. This will be complemented by the development of an appropriate web interface that will give access to part of the results to facilitate the users (industrial companies, social actors, policy makers, citizens to these information). In particular, the following methodologies are applied:

- **Evaluation of social impact of trade and new supply chain models**

The aim of this part of the project is to assess and quantify the socio-economic impact of the previously identified scenarios of current and future trends and changes in international trade and SCs. To this end, in a first step, data on global trade patterns, such as Global Value Chain data, trade flows and the RWI's Container Throughput Index, are analyzed to provide a quantitative overview of recent developments and the current state of global trade patterns. For this analysis, the project team can draw on its long-term experience and ongoing work in analyzing and forecasting (macro-) economic developments in Germany, Europe, and the global economy. One focus of the analysis can highlight developments during the recent global recession caused by the pandemic and the ongoing economic recovery. This will show how the crisis and the economic recovery are characterized by a substantial heterogeneity regarding the extent to which different countries, different economic sectors and different parts of the population are affected. In addition, an evaluation of world input-output tables gives insights into the importance of specific economic sectors in different countries and their global interlinkages. The results of this analysis provide a quantitative background for the further steps.

The main element of this work package is to assess the effects of the trends identified in the other work packages and the resulting changes in global SCs and trade patterns on socio-economic outcomes, such as GDP, employment, job quality, wages, and income inequality. For this purpose, a macroeconomic multi-country multi-sector network model will be set up, that comprises several economic sectors and sector specific trade flows. For each country, the sectors are represented by a production function that uses capital and labor as well as intermediaries as input factors. Intermediaries are a bundle of inputs potentially from all other sectors and all countries. The level of aggregation depends on the findings of the previous analyses. Moreover, the model includes different groups of households with different levels of labor quality, which leads to a variation of productivity. This heterogeneity in the productivity of households that lead to differences in labor income allows us to analyze the effects of SC changes on income distribution as well.

The main source to quantify the international value chains is the World Input-Output database. In addition, we use WIOD socio-economic accounts data that include employment, capital stock, gross output and value added at the industry level to set up the sectoral production functions. A simple way to identify different levels of labor productivity is to assume that they are concentrated in different industries. During the project we test different strategies to distinguish labor productivities.

To build this model, we can draw on several elements from existing models at RWI. This includes the RWI's existing macroeconomic multi-country model, which is part of the RWI's tool set for macroeconomic analysis and considers the external economic interdependencies of 39 countries, which are explicitly modelled, and the rest of the world. The core of the model is a trade matrix which maps the bilateral trade flows of the included countries. However, to account for global value chains we have to disaggregate international trade flows. On a macroeconomic level, the results from this model will show, how the changes in trade and production patterns will affect the individual economies in the different dimensions.

In addition, to get the appropriate level of disaggregation at the sectoral level we have to adjust the supply side of the country models. Moreover, we need to modify the labor market module of our model to take into account the different productivities of households. Recently, we have used a similar model to analyze the effects of a minimum wage in Germany. With the help of this model, the effects of the changes in global SCs on sectoral value added in the countries of the European Union and important trading partners can be analyzed. From this, effects on labor demand by skill

level and the income distribution can also be derived. This will provide additional insights on the challenges that might result from changing global SCs for different regions and parts of the labor force.

- **Social accounting for sustainability**

The methodology relies on both analytic and synthetic approaches. The first one is based on the social accountability through which the aggregated, traction and induced value are calculated for the whole value chain attending to the different stakeholders (customers, suppliers, employees, society…). In order to measure the social impact of the global value chain, primary and secondary data (using ORBIS database) will be used for developing a fuzzy logic algorithm which integrates the social impact of each of its links. This quantitative approach is accompanied by qualitative based methods to build a more precise model. The social impact will be therefore analysed in order to measure whether the impact is balanced between the different stakeholders. The synthetic approach refers to the aim of developing a compounded social impact measure. It also leads to represent all the results using graphs and matrices in order to make easier the interpretation of the results. This methodology aims to develop knowledge regarding the social accountability for its application in the entire value chains. It will also allow the development of a "selfassessment tool" for measuring the social impact of the entire value chain.

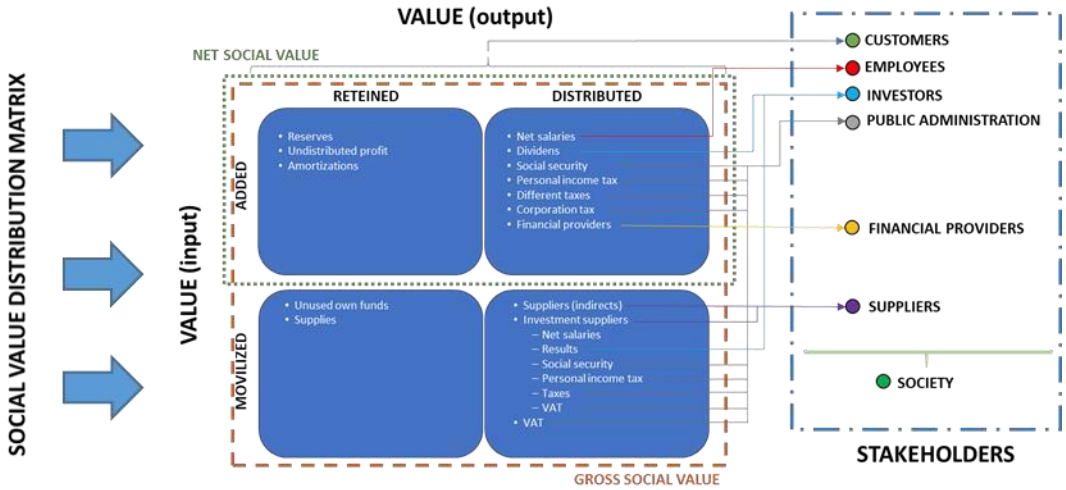

Fig. 1.4: Framework for the social accountability modelling

Obj. 4 Policy scenario building

The overall objective of the policy scenario building is to propose policy recommendations, whose consideration and implementation can support public and private organisations in overcoming economical, ecological, and social impacts from changing European SCs, e.g. due to the delivery of the European Green Deal, and to realise the European Pillar of Social Rights Action Plan. Figure 1 illustrates the methodology towards this objective.

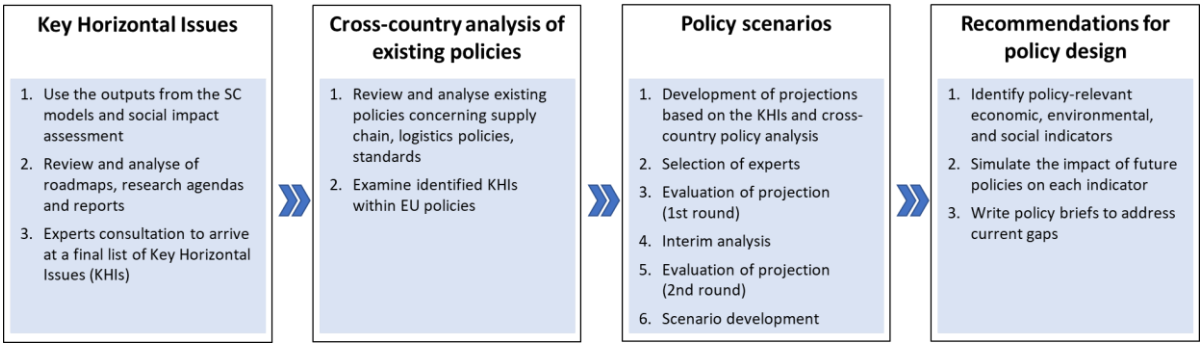

Figure 1.5: Methodology for Policy scenario building

First, ReSChape identifies and clusters Key Horizontal Issues (KHIs). This is because the scenario building process can be defined in four stages (Kosow et al., 2008): scope definition, key factor identification, key factor analysis, and scenario generation. However, the scope of the scenario is already defined in this proposal, this is, the impact of changing SCs and trade flows on economic, environmental, and social sustainability.

KHIs are issues that cut across sectors and confront most companies – both in the production and distribution sectors – like policies, education and capabilities, as well as demographic change. Therefore, they can be considered "horizontal". For the initial identification of the KHIs, both primary and secondary sources are taken into account. On the one hand, the findings from the SC models qualitatively developed in WP 2 and quantitatively evaluated in WP 3 serve as primary sources. The respective output on social indicators can be used as input on challenges and horizontal issues for this task. On the other hand, primary data will be collected through the consultation of experts from industry, research institutions, associations, and government agencies. The experts' consultation will be guided by scientifically sound guidelines, for example those of Morgan (2017)[1]. As secondary sources, roadmaps, research plans, research agendas, and reports that relate to KHIs and/or the economical, ecological, and social impacts of changes at the SC level will be considered.

The information collected from primary and secondary sources is then cross-checked and consolidated, resulting in a compiled list of validated KHIs. To differentiate between KHIs on a macroeconomic and on microeconomic levels, the identified KHIs will also be clustered. The use of a weighted, static design structure matrix according to Browning[2] is a suitable tool for this purpose.

In the next step, the key horizontal issues factors will be identified and mapped to the relevant public policies based on existing literature and interactions with stakeholders. The analysis of the impact of such policies on the key factors will be performed via simulations, using decision analysis methods such as cost-benefit analysis or multi-criteria decision models [3]. For instance, an analytics tool for assessing and monitoring the level playing field developments in key policy areas will be designed. This will be done by building a data driven model/simulation whose inputs will be macro-economic indicators (EU wide and country by country) for indicators linked for example to workers' rights (e. g. minimum wage, unemployment benefit), competition and state aid (or subsidies for business), taxation, social, and environmental protection. The model will aggregate data from different sources (e. g. publicly available databases as well as paywall/registration-wall information and EC online databases). The system can be developed using statistical and machine learning (AI) techniques in c++ and in Python notebooks (using libraries such as numpy, scikit, etc.).

The aimed policy scenarios will then preferably be generated in collaboration with all relevant stakeholders [4]. For this reason we will organize roadmapping workshops [4] to generate possible alternatives for the future. Therefore, we will discuss the results of our previous analyses in these workshops and will identify the most desirable set of policies which may help public and private organisations to overcome the diverse economical, ecological, and social impacts from changing European SCs within the respective scenarios. Using these scenarios as a reference, we will analyse which policies are best suited to tackle the current gaps in national in regional policies, and propose consistent policy mixes to tackle Europe's needs.

Networking with social, industrial and policy stakeholders

The implementation of the methodology is grounded on the capability of partners to create appropriate networks with social, industrial and policy stakeholders and involved them along the project. Partners have direct contacts from previous projects with several organisations that can be interested to contribute to the project participating to workshops, seminars, focus groups, survey and interviews. A specific mechanism to involve them will be put in place to create a network that can be activated along the 3 years of the project according to the specific needs and ongoing developments.

Networking social, industrial and policy stakeholders of the project will bring together a diverse set of actors with a collective shared interest in the achievement of value chain-related missions. These actors will give highly valuable

[1] Morgan, G. (2017): Theory and practice in policy analysis - including applications in science and technology. [https://doi.org/10.1017/9781316882665]

[2] Browning, T. R. (2001): Applying the Design Structure Matrix to System Decomposition and Integration Problems: A Review and New Directions. IEEE Transactions on Engineering Management, Vol. 48, No. 3, pp. 292-306

[3] Belton, V., & Stewart, T. (2002). Multiple Criteria Decision Analysis: An Integrated Approach. Springer Science & Business Media. [4] Guaita García, N., Martínez Fernández, J., & Fitz, C. (2020). Environmental Scenario Analysis on Natural and Social-Ecological Systems: A Review of Methods, Approaches and Applications. *Sustainability*, *12*(18), 7542.

[4] Siebelink, R., Halman, J. I. M., & Hofman, E. (2016). Scenario-Driven Roadmapping to cope with uncertainty: Its application in the construction industry. *Technological Forecasting and Social Change*, *110*, 226–238

contribution to the development of project results thanks to the many are not willing or able to engage in entrepreneurial activities associated with the ongoing development, deployment and scaling of such solutions.

We plan to create a group with external experts that cover an array of backgrounds: business owners, investors, customers, consumers, consultants and tech experts. A chairperson will be established who will undertake the management of the group and will involve them according to the planned activities. They will require: i) Motivation to create a systemic impact across value chains; ii) Specific knowledge of respective value chains, and of key technologies; iii) Knowledge of social dimensions and how companies are implementing specific actions; iv) knowledge on the role of policy supporting companies in implementing new social solutions.

Table 1.2 List of current contacts of ReSChape partners

Community /link	Members' Profile	# Members
Aspen Italia https://www.aspeninstitute.it/en	Entrepreneurial, political and social leaders, to promote the free exchange of ideas on challenges facing today's decision-makers in politics, economics, academia and the media, with particular attention to the business world and impact on society .	+300
Confindustria https://www.confindustria.it/en	National Trade Union in Italy, representing manufacturing sector. Ongoing collaboration with the R&D department and with several members.	+10,000
CILT (Chartered Institute of Logistics and Transport, West Midlands) https://ciltuk.org.uk/	CILT is a chartered body for professionals involved in the movement of goods and people and their associated SCs.	over 30,000 members in over 30 countries.
Lantegi Batuak https://www.lantegibatuak.eus/	Lantegi Batuak is a non-profit organisation whose aim is to promote and achieve social integration and employment for people with disabilities.	3000 employees
Siemens Gamesa https://www.siemensgamesa.com	Siemens Gamesa is a leader in the renewable energy industry, working to provide the world's best offshore and onshore wind turbines and services	24k employees
COVAP https://www.covap.es/	Agri-food cooperative	2,500 members
Provincial council of Bizkaia https://web.bizkaia.eus/	Provincial council of Bizkaia is the government of the Historical Territory of Bizkaia	Citizens, companies and associations of the Bizkaia region
CSCMP www.cscmp.org	Council of Supply Chain Management Professionals	500+
DIHNET.EU: Digital Innovation Hubs Community https://dihnet.eu/	Research institutions, industrial-technological centers, industrial associations, incubators, science and technology parks, industrial companies, technology companies, training institutions, and government agencies.	40,000

EIT Manufacturing AGORA: https://agora-eitmanufacturing.eu/	Social network and open innovation platform for the panEU manufacturing community. Manufacturing companies, technological companies, start-ups, researchers, universities, research and technology organizations.	500+
EFFRA - European Factories of the Future Research Association www.effra.eu	EU PPP involving companies from discrete Manufacturing sector, technology providers, start-ups, RTD organizations.	500+
ALICE - European Technology Platform in Logistics https://www.etp-logistics.eu/	The European Technology Platform ALICE is set-up to develop a comprehensive strategy for research, innovation and market deployment of logistics and supply chain management innovation in Europe.	150+
IG Metall (Metal Workers' Industrial Union)	employees from the areas of metals and electricals, iron and steel, textiles and clothing, wood and plastics, crafts	2.2 million +
https://www.igmetall.de/ueberuns/ig-metall--a-strong-community	and services and information and communication technology	
BVL (Federal Logistics Association) https://www.bvl.de/en	Cross-sectoral, international network of experts with specialists and managers from industry, trade, services and science	10,500 +
BVMW (Federal Association of Medium-Sized Businesses) https://www.bvmw.de/?lang=en	Cross-professional and cross-sector members of the small and medium-sized business community in Germany	55,000

1.2.2 Synergies with other national and international initiatives

Table 1.3 presents the synergies and progresses of ReSChape in relation to other projects where partners are currently involved or just finished. Cluster activities and cross-projects cooperation are expected based on the synergies identified in order to boost the impact of the actions and to make their results feed into ReSChape.

Table 1.3 Synergies with other projects and initiatives

Project	Synergies with ReSChape project
Next-Net (H2020 project-NMPB) https://nextnetproject.eu/	Next Net identified a set of SC strategies to face future challenges and related research and innovation priorities. ReSChape will exploit this work and further explore some of these strategies (like resource-efficient, closed-loop, human centered), analysing the related research and innovation priorities to evaluate which can be developed with a social perspective taking into consideration SC configuration and processes.

Untangled - (H2020-Transformation) https://projectuntangled.eu/	Untangled - Disentangle the effects of technological change, globalization and demographic change to promote shared prosperity in Europe ReSChape will build upon the output of this project giving a specific insight on the role of SCs in building a new social impact. In particular we will build upon: comparative analysis of the separate and combined effects of technological change, globalization, and demographic change on employability, skills, wage and income inequality, and labor mobility and migration in the EU; multi-faceted scenarios of technological progress and globalization and their impact on national, regional, and sectoral labor markets and incomes in the EU; policies that are conducive to shared prosperity by helping to reap the benefits of these megatrends and mitigate their negative effects.
Circular 4.0 (Alpine space project) https://www.alpine-space.eu/projects/circular4_0/en/home	CIRCULAR4.0 will increase the knowledge of Entrepreneurs and Intermediaries on the circular economy concepts, develop and test a set of tools able to foster the contribute of Industry 4.0 for the transition of SMEs to the circular economy. With this project, ReSChape will share how to address Policy Makers, Intermediaries and SMEs to raise the importance of digitalization processes, focusing mainly on: the innovation processes, the business models and the investments needed.
DigiCirc	DigiCirc is a cluster oriented project designed to meet the targets and ambition set out by EU Circular Economy policies and actualise circular value chains as a competitive advantage. DigiCirc empowers SMEs to leverage digital technology as a key enabler for innovative circular products/services, processes and business models. ReSChape will further explore new business models and implement a social sustainability plan towards fundamentally restructuring value chains to be low-carbon and circular. https://digicirc.eu
Vanguard initiative	Network of regions focusing in advanced manufacturing and with a special emphasis in circular manufacturing. https://www.s3vanguardinitiative.eu/
SORISMA	In the SORISMA project, tools are developed with which companies can manage the risks arising in the three dimensions of people, organisation and technology in the context of Industry 4.0 projects. e. g., methods for risk analysis or the development of risk strategies enable companies to identify the effects and risks of Industry 4.0 and to derive suitable measures to minimise risks and this results can be input for ReSChape for the analysis of risks due to disruptions.
Center of Excellence Logistics and IT	The Center of Excellence Logistics and IT forms an ecosystem for logistics innovation and supports the transformation into new organizational forms such as an inter-company social networked industry where focus is on people as an essential element.
Innovationlab Hybrid Services in Logistics	The focal point of the Innovationlab's work lies in the question of how to design and organise responsible and target-oriented ways for people and technology to interact in shared networks. Therefore, innovative technologies are being developed and tested in cooperation with companies while focusing on the social question of how to design and pay for work 4.0

Siemens Gamesa's contribution to the SDGs	This project is based on the analysis and monetization of the social value that Siemens Gamesa's social action projects generate for the different stakeholders. The value contributed for each SDG in which the company operates is quantified. Recommendations are enounced for the selection of projects and proposals for optimization in the creation and distribution of the value generated, based on equity among stakeholders and their contribution to the SDGs.
Social value generated and distributed by the Port of Cartagena	The objective is the development of a model that allows systematizing the social accounting of the Port of Cartagena, identifying and promoting the Market, non-Market, induced and emotional value that generates the different interest groups and its repercussion on the environment.
AgriCoopValue. Erasmus K plus Proyect (2021-22)	The aim of the project is the systematization of a methodological process that allows social accounting to be scaled to the group of agri-food cooperatives of the European Union, based on the experiences developed by the Spanish territorial entities of social economy.
FOODRUS Circular solutions for resilient food systems	The FOODRUS project seeks to create resilient local food systems in Europe by developing 23 circular solutions to limit food losses and waste along the agri-food value chain. It will promote efficiency in the use of resources across all the stages of the value chain, through process optimisation and sustainable dietary and consumption toolkits.

1.2.3 Interdisciplinary approach

ReSChape project has a highly interdisciplinary nature, reflected in the composition of the consortium – which encompasses institutions of different types involving professionals and researchers with varied backgrounds and from different disciplines (such as economists, management engineers, industrial engineers, social policy scientists). Due to the complex framework of the project, as well as the comprehensive features related to the need to build a social resilient and sustainable Europe, we believe that the involvement of different disciplines is essential to the success of the research both with regard to its theoretical contribution and practical range. Moreover, the project is ambitious in terms of interdisciplinarity as it will boost the collaboration between different contexts, such as: industry (key value chains), policy makers, academics, social entrepreneurs and technology professionals.

In terms of expertise and methods, the joint effort of partners from different disciplines will enrich the results of the project as we understand that, when well-managed, multidisciplinary teams can perform better on tasks that benefit from multiple perspectives than homogeneous teams. As creativity is a fundamental part of innovation it will be stimulated during all the phases of the project, the collaboration and the sharing of ideas within the group – which generate novel associations and can lead to additional ideas – will be continuously encouraged. A social innovation approach will be used as it systematically addresses two of the main challenges faced by European companies and value chains today - their resilience and their sustainability.

1.2.4 Integration of social sciences and humanities

Social sciences and humanities play a fundamental role in the ReSChape project, as they are key to understanding social issues, such as the balance between economic development and the impact of industry on the social and environmental dimensions. As mentioned in section 1.2.3, both the composition of the consortium and network of experts to be created will encompass institutions, professionals and academics from a variety of disciplines, including social sciences and humanities. The role of social sciences and humanities in the project starts with the overall identification of the social changes and expected impacts that will be addressed along the project and continue in all tasks of the project. The definition of the principles of social resilient and sustainable value chains and the principles for social SC have a high contribution from social sciences. Professionals and institutions from the following social sciences and humanities disciplines are expected to be involved during the different phases of the project: (i) Social and behavioural sciences: economics, political science, sociology, human rights; (ii) Journalism and information:

journalism, archival sciences; (iii) Business and administration: marketing, management, public and institutional administration; (iv) Humanities: ethics.

1.2.5 Gender dimension

ReSChape partners understand gender equality as "an essential condition for an innovative, competitive and thriving European economy" as outlined in the European Commission Gender Equality Strategy for 2020-2025 - A Union of Equality, and are committed to fostering gender equality in research.

Partners will strive for a gender balance throughout all project phases, starting with partner organizations and staff. Among the key staff listed in the participant profiles both male and female key persons are listed and, during the project, active measures will be taken to ensure that women scientists and practitioners are well represented in the partners' organizations and project leadership. Partner organizations will take measures to ensure accessibility of meetings and activities for all. Continuous monitoring will be implemented and all period reports will include a section mentioning the number of men/women working on the Action per beneficiary.

Gender equality guidelines and targets will also be implemented during the project phases that include the participation of external stakeholders, and gender indicators will be considered when evaluating the impact on social dimension of decisions related to SCs and to policy.

1.2.6 Open science practices

The project will apply an open and collaborative approach based on systematic sharing of knowledge developed in the project. A balanced approach between the adoption of Open Science practices and the assurance of the Intellectual Property Rights of all stakeholders will be adopted in the ReSChape project, aligned with the EU data sharing policy on the principle "as open as possible, as closed as necessary". The partners understand intellectual property as a crucial aspect when it comes to foster knowledge co-creation and are fully committed to ensure its accomplishment. Open science will be implemented as an integral part of the project by means of three main practices that help to improve the transparency and accessibility of the scientific research and, as consequence, increase both its validity and recognition within the academic community and its social contribution:

- Preregistration – where possible, the hypotheses and objectives that arise from the research in the context of the project will be developed and published before their test as a way to assure the transparency of the process. Similarly, all the methodological procedures, including the data analysis plan, will be publicly registered prior to conducting the study. These practices help to ensure that the objectives and methods are not changed along the research. An online platform will be used to preregister the research intentions (e.g. https://aspredicted.org/).
- Open data – open data will be considered an alternative during the project when applicable, assuring the Intellectual Property Rights of all stakeholders and considering data ownership issues.
- Open access – preference will be given to publish the results of research in the context of the project with open access. Pre-prints will also be considered as an option to publish results.

1.2.7 Data management policy

ReSChape project work is based on data collected from companies, and in general information collected from stakeholders like social actors, policy makers, industrial companies. These data are related to business and no personal data need to be collected for this project. Moreover, data from public databases (like Eurostat, OCSE, World Bank, national statistics, databases with accounting information like AIDA in Italy) will be used along the project for the modelling of social impact. The project deliverables related to project coordination (WP1) will include a detailed data management plan (DMP-D1.3) concerning the FAIR principles related to all collected data. In general, data collected will be stored under the Project Coordinator's private database with limited access to the consortium members, in line with European GDPR (UE/2016/679). The database will be maintained under supervision of the Project Coordinator's data management officer. Data produced by the demonstrators is either owned by the individual applicants of each demonstrator or, collectively, by ReSChape consortium, in which case it will follow the defined guidelines of the DMP. A Data Manager will be identified, ensuring that participants are provided with relevant guidelines and documentation for data management. Where applicable, metadata structure and formatting will follow standard practices to ensure replicability and anonymity, following Intellectual Property Rights and data ownership concerns of involved stakeholders.

2. Impact

2.1. Project's pathways towards impact

2.1.1. Outcomes and wider impacts

Outcome 1 - *Develop knowledge on the ongoing and expected changes and disruptions in trade patterns, global value chains and production networks*
ReSChape contribution: identify and analyse social, economic and environmental impact of contemporary issues on trade patterns (obj.1), global value chains and production networks, including the causes and effects of disruptions (obj.2). On the basis of the results, ReSChape will propose new SC models for more resilient and sustainable value chains. **Target Groups:** industrial business owners, supply chain managers, technological companies, industrial associations, policy makers, researchers. **Scientific impact**: theory building on new models for resilient and sustainable value chains applying different framework of analysis like complexity and contingency theory. **Economic/technological impact:** The knowledge developed in ReSCape on the expected changes in global value chains and the post pandemic SC models, will create better understanding among industry and policy makers on how to push forward the drivers of change to achieve the long-term goal of digital technology adoption of at least 20%. Moreover, the human-centric approach is expected to increase human-machine interaction level in manufacturing and logistics processes by 30% by 2030. **Societal impact:** ReSChape contributes to enhance employment and social fairness by designing post pandemic adapted SC strategies, such as resource efficient, circular and human centered value chains. **Requirements and barriers:** to accomplish this outcome, ReSChape will need to map the contemporary challenges faced by companies and value chains and to develop models that contribute both to theory and practice when it comes to more resilient and sustainable value chains. The main barrier is the complexity associated with the analysis of the changes in post pandemic global value chains. **Scale and significance:** ReSChape intends to start from the trends mapped in NEXT-NET and to update them with a specific focus on social trends linking to the 20 EU pillar of Social Rights. The project intends to engage 100-150 companies during the development of 4-5 models for resilient and sustainable value chains.
Outcome 2 - *Identify innovative ways to maximise the potential and mitigate the adverse social, economic and environmental impacts of changes in global value chains and international trade patterns in European urban and rural areas. Research should take into consideration impacts on employment, job quality, economic growth, income inequalities and on social cohesion and well-being*
ReSChape contribution: definition of innovative SC models (obj.2) that mitigate the adverse social impacts of the COVID-19 pandemic, definition of innovative systems to assess social impact (obj.3), towards supporting informed decisions about configuration of new SC relationships and collaboration across regions

and countries for new global trade patterns. Moreover, we will provide analysis to link skill gaps with employment indicators.

Target Groups: industry, academia and government

Scientific impact: Development of theoretical models for supporting evaluation of social impact in SC literature. The results can support also universities wishing to gain an understanding of the related skills contributing to the success of its graduates and any skill gaps that may exist in its current curriculum i.e., design new education programs, and redesign and upgrade the existing program.

Economic/technological impact: Enabling companies to value intangible assets to calculate the net social value of their SCs. Helping companies to widen the CSR framework and to automatise the collection of information on impact to save time and increase reputation.

Societal impact: More transparency on business practices related to employment, job quality, well-being of workers. Moreover, for what concerns skills, this project will propose answer to unemployment issues to overcome skill mismatch proposing solutions regarding different sectors taking into consideration urban and rural conditions. As an example, the project may impact positively on youth employment which was up to 16.8% in June 2020 in EU, and in some regions the situation was worse e.g. 40.8% in Spain (Kraatz, 2020).

Requirements and potential barriers: the project will need to engage a set of different stakeholders and experts from various fields will need to be able to collect a variety of data. The difficulty to access employers to collect the data is a potential barrier. Secondary data e.g., jobs ads could be used.

Scale and significance: These outcomes will be a valuable contribution to help companies and governments in instantiating the ambitious goals of the 20 EU Pillars on Social rights.

Outcome 3 - *Assess the impacts of the coronavirus pandemic and related public health crisis on global value chains, production networks and security of supplies, and their short- and longer-term implications for employment and social resilience in the EU*

ReSChape contribution: Support the decision making in organisations to face the unintended and indirect consequences of the COVID-19 outbreak and the related SC strategies and risk management responses of the manufacturing companies (obj.3). Support policy makers in the design of policies to overcome the impacts of the coronavirus pandemic (obj. 4).

Target Groups: Manufacturing companies and policy makers

Scientific impact: Provides new knowledge that can be used by researchers and managers to resolve future SC risk challenges and avoid the need to relearn lessons already learned by COVID-19 research when the next global disruption comes around.

Economic/technological impact: on the models proposed by the project, companies can enhance their resilience and achieve competitiveness. Social accounting facilitates the identification of the displacements that have occurred due to the COVID-19 such as gaps in the generation or distribution of the value generated related to the covid data (by years and countries). A social value generated self-assessment tool leads the strategies of the companies to understand the unbalanced impacts depending on the stakeholders that are considered. This fact makes companies to re-think their value chain model, thinking on response, reactive and recovery measures.

Societal impact: Facilitate public bodies in monitoring long-term consequences (e.g. youth unemployment, diversity, excessive working time) of disruptions caused by COVID and establishing appropriate corrective measures

Requirements and potential barriers: Due to the contingent situation, it might be that managers do not share the information about the strategies they are putting in place due to confidentiality and related competitive advantage. Difficulties in collecting fresh and updated data on the ongoing situation.

Scale and significance: The cross-country analysis will provide insights not only on the impacts of the coronavirus pandemic, but also on the results from the various policy measures put forward by the European governments. .

Outcome 4 - *Unfold policy scenarios for future value chain developments, production networks and level playing field trade relations, which will ensure security of strategic supplies, strengthen the economic resilience of societies, foster sustainable employment creation in the EU and mitigate the impacts of future trade and value chains disruptions on EU employment*

ReSChape contribution: Questions asked about the preparedness or even existence of policies and associated systems to help individuals and communities in EU in the field to deal with the consequences of COVID-19 and other key megatrends. Thus, this project will fill this gap by providing policy recommendations for future

value chains.

Target Groups: policy makers, researchers

Scientific impact: The scenarios will consider the resilience and sustainability (in all its facets) of European SCs in the context of global value chains.

Economic/technological impact: Testing the possibility of using a comprehensive social accounting of the value chain will enable three complementary outcomes; (1) To identify the information (transparency / information symmetry) that would be necessary to demand from companies in order to establish value chain social accounting; (2) This accounting information, combined with the Morphological Analysis, would allow us to establish prospective analysis scenarios for suggesting new or redefined public policies. (3) Complementarily, from the mentioned Morphological Analysis it would be possible to infer the potential impact of the various countries (territories) involved in the value chain, allowing progress in cooperation models, both public-private, and between countries with complementarity in the value chain.

Societal impact: Government regulations impact every SC, and companies should adapt their operations to overcome any trade barriers in terms of expertise, inputs and technology, which will have negative effects on employment, economic growth and sustainability.

Requirements and potential barriers: From a visionary point of view, partners and stakeholders may have a limited vision of the future with little or no capability to see forward the future policies. Secondary data e.g. reports also will help us to capture and verify our results.

Scale and significance: ReSChape will map and identify 6-7 Key Horizontal Issues related for example to trade conditions, taxation, international agreements, and define 3-4 policy scenarios. Simulation and what if analysis should bring to an index of consistency of the retained scenarios for over 90%.

Outcome 5 - *Produce new, innovative methodologies and ways for assessing and monitoring the level playing field developments in trade and value chains in the EU*

ReSChape contribution: analysing trends and changes will include also monitoring the level playing field, and assessing the impact of trade on social indicators, the societal impact through social accounting for sustainability method, and the economic and environmental impacts within industrial business concerning major issues on cross-country EU policies.
Target Groups: policy makers, multi-national companies, researchers and academia
Scientific impact: The innovative methodologies will provide new knowledge on assessing the impact of different kinds of trade and value chain disturbances. Policy makers can use them to improve decision making processes. Researchers can also employ the ReSChape results to increase the number of empirical cases for continuing developing this knowledge area. These new methodologies will enable practitioners and academics to a better understanding about the social value generated and its impact on stakeholders and social well-being.
Economic/technological impact: Enabling better assessment and monitoring of trade conditions (i.e. taxes, partnerships between countries, custom duties, etc.) in order to contribute to facilitating fair competition and equal opportunities to countries and companies across different EU sectors. On the technological perspective, the faster and easier diffusion of novel methodologies and technological advances throughout different EU sectors and countries will enable a "fair play" perspective in terms of competition and knowledge sharing. In the long term this will result in fairer trade conditions, as well as technological competitive advantages, for EU compared to non-EU countries.
Societal impact: Facilitate implementation of rules and standards to prevent businesses undercutting rivals in areas such as workers' rights and environmental protections. Companies, non-profit organisations, public administration will be provided with feedback on their social impact which will enable them to increase their legitimacy in the society and as consequence the citizens and the regions will be benefited.

Requirements and potential barriers: capability to involve EU policy makers and support them to deploy the suggestions and indications for new regulations and directives on trade rules and new social accounting. Non-EU countries do not accept proposals from EU on level playing field. Industrial partners can be refrained from sharing knowledge with peers, as well as providing insights into technological diffusion on a sector scale.

Scale and significance: Potentially the scale is very large with impact on trade (increase of export of EU goods, reduce difference in taxation to business among countries) and with ripple effect scaled upstream regarding economic and social impacts on the value chains**.** Increase 20% of export of EU goods in 5 years, reduce difference in taxation to business among EU countries, increase diffusion of innovative technologies and knowledge sharing for competitive advantages in trade within EU countries.

Long-term impact for destination: *Social and economic resilience and sustainability are strengthened through a better understanding of the social, ethical, political and economic impacts of drivers of change (such as technology, globalisation, demographics, mobility and migration) and their interplay.*

ReSChaPe pathway to contribute to the targeted expected impact of HEU Strategic Plan:
ReSChape directly and effectively contributes to this long-term impact fostering the transformation of present SCs and production networks towards a new approach where social and economic resilience and sustainability are reached using methods and digital tools to allow stakeholders to know which are the drivers of change and to have the capability to answer. The developed tools will study the interplay between dimensions (social, economic and environmental) both at macro and micro-level. The project will refer to the UN-SDGs as well as to the 20 European Pillar for Social Rights and will define how the different involved stakeholders can support their implementation.

2.2. Measures to maximise impact - Dissemination, exploitation and communication

2.2.1. Dissemination and Communication (D&C)

The objectives of the D&C and exploitation strategy are:

- To network relevant stakeholders for consultation process and to keep them updated on the most important findings of the project;
- To support the validation of project results with involvement of different stakeholders;
- To organize meetings with policy makers to support European Commission in defining new policy scenarios;
- To promote knowledge created in the project through actions for a wide audience and to increase the visibility and consciousness on the importance of the research on SC topics and policy scenarios towards a wide audience

- To exploit project results in the European SC Observatory which would represent a platform where to share knowledge with stakeholders about SC.

In particular, the project partners strongly believe that in order to optimize dissemination and communication actions, it is crucial to make concrete showcases of project outcomes and knowledge available to an increasingly expanding audience. This includes networking partners and other stakeholders not included in this project. The ReSChape project therefore takes an iterative stakeholder expansion strategy. In case of results that cannot be shared due to confidentiality issues, they will be presented in an adapted form for all target groups. Relevant project materials will be available through the project website and will be published through the project communication channels including newsletters and other public material. D&C manager, together with the Project partners will organise the dissemination and communication so that maximum reach is obtained, at the same time controlling licensing and IPR issues related to content. The Dissemination strategy consist of three phases: *a. Dissemination phase 1 – inwards oriented*

- Validate the ReSChape knowledge base and solutions in a contained set of stakeholders, expert groups with a limited (e.g. 6-10) industrial and academic organisations in Europe using good practices and tools from previous practices from a European wide network of cross-sectorial SCs. In fact in the first phase of the project, a European wide study will be done with 50 – 100 relevant organisations to collect relevant good practices related to projects on SC scenarios and that can help to define the future scenarios.
- In this phase dissemination, focus will be on general communication about the project to the general public (e.g. through website) and making knowledge, and good practices available to facilitate the smaller initial group of stakeholders.

b. Dissemination phase 2 – intra-clusters oriented

- Expand the number of industrial stakeholders involved in the process, and validate the ReSChape Project future scenarios for SC and research and innovation priorities on a larger scale, involving (an estimated 30 – 50) additional stakeholders in the ReSChape selected sectors.
- In this phase, dissemination activities will focus on fine-tuning the knowledge base and research needs on the results of the project, and making them available in a user friendly way to a wider audience between the different clusters.

c. Exploitation and dissemination phase 3 – externally oriented

- Expand the audience of the ReSChape project to a much wider audience of related stakeholders (500 – 3000) throughout Europe, fostering take-up of ReSChape findings and results for their implementation targeting a vibrant interconnected European network of cross-sectorial industry.
- Dissemination activities will be focused on further professionalising the knowledge, and making sure a wider audience of relevant stakeholders has appropriate knowledge about the ReSChape project and its results, in order to stimulate their take-up and accelerate resource efficiency.
- Exploitation activities will be targeted at developing an “exit strategy” or a “sustainability strategy” for the joint ReSChape results, investigating how the ReSChape can be continued and financed beyond the end of the project.

In order to properly execute and monitor the dissemination and exploitation activities, a detailed plan will be made at month 12 of the ReSChape project. This plan will not only include the more detailed definition of deliverables and any additional activity required, but also refine the approach to delivery outlined here. On the following pages, the preliminary dissemination, communication and exploitation activities for ReSChape are reflected.

Dissemination and communication related to promoting the ReSChape project and its results will be implemented through strategic and targeted measures possibly engaging in a two-way exchange with the target groups. The target groups will receive easily understandable information on technical project results. The dissemination and communication plan will apply diverse information regimes to different stakeholders, based on the importance they have for the project and their interest in the outcomes of ReSChape:

- **Players with high interest/high influence**: have a high interest in ReSChape and have a high potential influence on the project. These players will be engaged closely to build a relationship during and after the project (e.g. Industry, policymakers).
- **Players with high interest/low influence**: have a high interest in ReSChape, but do not have a high (beneficial) influence on the project. These organisations will be regularly informed in order to keep their interest.

- **Players with low interest/high influence**: can have a high influence on the starting of the project (trend setting and analysing), but a fairly low interest in ReSChape results. Actions will be aimed at increasing their perceived interest in the project without informing them too frequently.
- **Players with low interest/low influence**: consumers and citizens shall be involved through specific associations, consumer goods association, will receive only specific information and might participate to some part of the validation process as they cannot play a decisive role for ReSChape.

2.2.2. Communication activities

In the beginning of the project, there will be only generic news to share through the mass media and social media platforms, and communication will be focused on making the general public aware of ReSChape and its potential role in assessing the needs for the scenario analyses. In this phase, communication activities will create a coherent identity for ReSChape, supported with audiovisual content (logo, promotional videos, leaflets, etc.) and will deploy generic communication measures, like the ReSChape website and periodic newsletters with generic information about the project and related news. The ReSChape website will include the general information about the project, news & events, download section and interactive section, that will offer the opportunity to engage the different audiences during the duration of the project. Through the website target audiences will be invited to play an active role in the process of collecting, reporting, analysing and disseminating content. Examples of possible interactive activities supported are: blog/videoblog, wikis, RSS, tagging and social bookmarking, music-photo-video sharing, and podcasts. Such e-services, will contribute to the involvement of end-users as active participants.

As ReSChape progresses, project results become available and more insights are obtained in which stakeholders should be more closely involved in the project. The information which becomes available provides the opportunity to involve external stakeholders into the project by changing their perceived interest and attitude in the project with targeted information. Strong media coverage will be another major target and media campaigns will be organized in relevant social networks to disseminate the project but also to raise awareness of the general public towards key messages of the project.

The communication strategy will be designed by ReSCHAPE communication manager, ZLC communications director, in charge of designing and supervising ReSCHAPE audiovisual content creation, graphic design materials, as well as communication content in general (such as press releases). She will also manage ReSCHAPE social media and organize ReSCHAPE events.

ReSCHAPE social media will include Twitter, LinkedIn and YouTube, and it will leverage social media allies to ensure maximum impact.

The expected outcome of ReSCHAPE communication strategy is presented in the tables below:

1) Media coverage and website expected KPIs

Media coverage	**ReSChape**	**Website**	**ReSChape**
Number of articles in local, regional and national media outlets echoing research work/events organised by the project (**clippings**)	20	Site **bounce** rates	8000
Press cuts' **reach**	30	Page **views**	15000
Tone of articles in local, regional and national media outlets echoing research work/events organised by the project	Positive	Unique page **visitors**	6000
Digital leaflet	1	External website **links**	10

2) Newstletter, events and videos expected KPIs

Newsletters	**ReSChape**	**Events**	**ReSChape**	**Videos**	**ReSChape**

Number and type of **subscribers**	100 subscribers (20 academia, 20 practitioner, 20 policymaker, 20 NGO, 20 other)	Number and type of targetted **participants**	250 (50 academia, 50 industry practitioners, 50 associations, 20 policymakers, 50 NGO, 30 other)	Number of **views**	200
Number of **downloads**	350	Media **coverage** targets	Printed and digital mass press, specialized magazines, radio, TV, Internet podcasts	Average **view time**	
Number of newsletters sent	6			Number of **shares**	20

3) Social media expected KPIs

Twitter	**ReSChape**	**LinkedIn**	**ReSChape**
Number of non-affiliated independent **followers** *versus* number of users followed	Up to 150 followers /up to 200 followed	Number of **followers**	Up to 500
Impressions (number of times your content is displayed, no matter if it was clicked or not, and therefore delivered to someone's feed)	Up to 50000	Posts	Up to 80
Reach (number of unique people who see your content) targets	Up to 300	Unique page visitors	Up to 800
		Page visualization	Up to1500

2.2.3. Dissemination activities

Dissemination in the context of ReSChape is regarded both as a information process targeting the the scientific community, specialist media, peers (e.g. other EU-funded projects), field actors (e.g. agencies, institutes, universities), as well as Programme Committee members, EU policy officers, among others, also with significant marketing initiatives aiming to promote the new SC concepts, future scenarios and business models to the business world and policy makers to influence the models of the future.

Dissemination activities will be implemented through a number of channels and a number of actions has been planned to assure the effectiveness of the general dissemination activities. Indeed, dissemination of project results through participation at international conferences, workshops, seminars, and publication in scientific journals will take place. All publications will be done with an open access status.

Scientific dissemination will be based on participation to recognized international conferences in the area of Supply chain Management, and generally Industrial Engineering, Social science and policy making. Time by time it will be evaluated the opportunity to make Special sessions within the conferences where partners participate. Results will be periodically published in scientific journals with international peer review and in the form of open access. Some journals that will be considered are: International Journal of Production Economics, Omega, Technology Forecasting and Societal Challenges, International Journal of Production Research. The partners will also discuss and evaluate the possibility to issue an open access Springer book with the most important results of the project.

Expected outcomes of ReSCHAPE dissemination activities beyond scientific publications include policy recommendations (leveraging WP4 deliverables as well issuing EC policy briefs), and industry-academia collaborations.

The following list in table 2.1 includes events and conferences where project partners usually participate and where possible dissemination activities can be realized.

Table 2.1 List of events for Dissemination

Event	Type of audience	Type of action	Expected impact
World Manufacturing Forum	high level policy makers and corporate leaders	roundtable and workshop	Potential audience: 500 people
Future Congress Logistics	3PL, distributors, logistics companies, manufacturers, research organisations	roundtable and workshop	Potential audience: around 300 visitors Workshops with 50 people
CSCMP annual conference (Council of Supply Chain Management Professionals)	SCM professionals (CSCMP annual event and/or CSCMP roundtables), research organisations (European Research Seminar organized in parallel to annual CSCMP event)	roundtable and workshop, scientific papers	Potential audience: Informed 200 people (for annual events) Roundtables with 50 people
Supply Chain Meeting 2017-2019 (Portuguese Annual Logistics Event)	Manufacturers, 3PL service providers, distributors, logistics companies, logistics equipment, ICT companies, research organisations	Event and roundtable discussion	Informed 1500 participants
Workshops with policy makers	Policy makers	Project results discussion, SRA for next Framework Programme	Potential participants: 100200 per workshop
EurOMA (European Operations Management Association) Conference	academics and practitioners	Project outputs, event, networking	Potential audience 400 participants
Logistics Research Network Conference	academics and practitioners	Project outputs, event, networking	Potential audience 100 participants
ISBEE World Congress - International Society of Business, Economics and	academics and practitioners	Project outputs, event, networking	Participants from more than 40 countries. Potential audience: 300 participants
Ethics			
International Congress of Spanish Accounting and Business Administration Association (AECA)	academics and practitioners	Project outputs, networking event, scientific paper	Potential audience: more than 300 participants
UPV/EHU Summer Courses Foundation	academics and practitioners	Summer course (June 2023)	Potential audience: 100 participants
International Supply Chain Conference	academics and practitioners	Project outputs, event and roundtable discussions, networking	Potential audience: Around 3,500 conference participants from over 40 countries

Hamburg International Conference of Logistics	academics and practitioners	Project outputs, event and roundtable discussions, networking	

2.2.4. Exploitation

Some of the partners of the project are working on a follow up action of the NEXT-NET project which is the creation of an European Observatory for SCs. The observatory aims to offer to its stakeholders updated information on the status of the SC in Europe by means of best practices, interactive access to indicators and benchmarking, projects. The observatory is a knowledge sharing area where academics and professionals as well as policy makers can find information and discuss about implementing different SC strategies for sustainability, resilience, agility, etc. In particular, part of the results of the ReSChape project will be exploited merging them in the platform integrating the missing dimension of the analysis of social dimension of SCs which is now missing in the overall concept of the Observatory (Fig. 2.1).

Figure 2.1 Exploitation towards the European Observatory for Supply Chains

Moreover, the results of the project will be used by partners for creating new training paths both for undergraduate and graduate students. Moreover, it will be useful for supporting consulting activities with companies and policy makers to help the identification of new paths for reaching some of the 20 European social pillars.

2.2.5. IPR management

Intellectual property (IP) management in ReSChape project is required to: 1) Protect IP made available by consortium partners and tools and methods used and developed during the project; and 2) to protect pre-existing IP made available and further developed.

IP management within the consortium will be defined in the Consortium Agreement (CA), in accordance with DESCA 2020 model, primarily, providing for the following:

- *Ownership and protection of foreground:* Foreground project assets will be owned and licensed in accordance with stipulations in the Consortium Agreement. This includes various project results developed by the consortium to be disseminated and exploited, as described in these respective sections. The project's opensource assets will be appropriately licensed as such (e.g. GNU GPL, Creative Commons, FreeBSD, Apache License v.2). Ownership will rest with the consortium partner that created or otherwise developed it. If more than one consortium partner has contributed and the individual contribution cannot be isolated from the rest, the foreground information will be owned jointly by contributing partners. For assets that are not open source, the owner will be entitled to grant an exclusive license to third parties. Procedures for conflict resolution management, should the need arise, will be specified in the CA.
- *Confidentiality:* Any confidential material provided/produced during the project will remain for the information of the project partners. Such information will not be disclosed to any other party without explicit authorisation.

- *Access rights to foreground:* Consortium partners will grant access rights to foreground information that is required to achieve the project objectives. The terms of access rights will be specified in the CA.
- *Access rights to background:* Consortium partners will grant access rights to background information that is required to achieve the project objectives, as detailed in the CA. Background rights will remain with the consortium partner that owns the rights.

For what concerns **IP management relating to ReSChape project beneficiaries** is a key focus of the project as a key element to unlock and spur exploitation/ market uptake, and directly relates to the project's Specific Objective ReSChape. The result of this will be that researchers will be empowered to: (i) take appropriate protection measures regarding their data, project results, and other assets (i.e. patents, proper and technical definition of know-how during licensing and other agreements); and (ii) gain knowledge of IP issues as they relate to possible negotiations with investors, large enterprises, and others to ensure that IP is properly valorised. Any IP generated by the partners shall rest with them. External parties involved in the project will sign Non-Disclosure Agreements indicating that they will not improperly make use of the information, technical designs, or other sensitive information of beneficiaries, made available to them through their involvement in the project.

2.3. Summary -KEY ELEMENT OF THE IMPACT SECTION

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
Secure autonomy of European value chains to avoid supply disruptions of key products, contributing to building resilient businesses; Improve the sustainability of European Industry, considering its environmental, social, and economic aspects. Need to reinforce the capability to assess social impact of business, giving a new meaning to social resilience. Need of new policies for business to implement social oriented actions.	Updated mapping of trends, changes and disruptions. Proposing SC models and innovative ways to organize SCs for social resilience Develop quantitative models for assessing and monitoring social indicators at multi-country and multi-sectoral level. Assess impact of COVID in the short and long term Propose policy scenarios for social resilience and sustainability.	(D) Dedicated initiatives: consortium meetings, workshops, public open initiatives to engage communities, webinars, scientific conference, final conference (C) Social channels (Linkedin, Twitter, Facebook, Instagram, and others) to reach broad audience (E/D) innovative models for social impact assessment for industry, policy makers, social actors for consultancy and training. (E/D) EU observatory on SC with a session dedicated to social impact and related policies (E/D) Open Access Positioning Papers
TARGET GROUPS	**OUTCOMES**	**IMPACTS**
Industrial companies, logistics providers, distributors, suppliers of important EUSCs like fashion, automotive, aerospace, food and others. Social actors including social unions, trade unions, social innovators, consumer associations. Policy makers both at regional, national and European level.	Awareness on the competitive advantage brought to SCs by a new approach to social impact. New social indicators used and monitored by companies at micro-level with a new system of social accountability New aggregated social indicators used and monitored by policy makers to evaluate the progress of the EU 20 targets on social rights and to implement innovative level playing field strategies. Users can access in interactive way to indicators by means of digital tools.	Scientific: New breakthrough scientific approaches on social impact of trade and SCs, on policy scenarios. Economic: help EU to reach targets of 75% of the population aged 20 to 64 in employment by 2030. Increase competitiveness of EU trade market. Societal: help implementation of better working conditions, decrease job disparities, increase balance in policies definition.

3. Quality and efficiency of the implementation

3.1. Work plan and resources

ReSChape activities are structured along 3 years along 5WPs which are interrelated each other according to the figure 3.2. While there is a WP for Project Management and one for Dissemination &Communication, there are 3 WPs with scientific activities namely: WP2 covering *obj.1* and *obj.2* thus identifies and analysing trends and changes and proposing related SCs strategies, WP3 covering *obj.3* proposing different approaches to analyse impact of trade and SCs on social indicators and finally WP4 covering *obj.4* developing policy scenarios.

Project GANTT

		WORKPLAN		START	END	DUR.
WP	1	**Project Management**	**CNR**	1	36	36
T	1.1	Consortium and project management	CNR	1	36	36
T	1.2	Quality Management	CNR	1	36	36
T	1.3	Data Management Plan	ZLC	1	36	36
WP	**2**	**Global trends and Innovative supply chain models**	**CNR**	1	18	18
T	2.1	Identification of trends, changes and disruptions	IML	1	6	6
T	2.2	Specific challenges for SC of the future	CNR	4	8	5
T	2.3	Definition of SC strategies for resilience and sustainability	INESC	6	12	7
T	2.4	Innovative ways to organise SCs	CNR	8	16	9
T	2.5	Skill and employment in logistics and supply chain	Aston	12	18	7
WP	**3**	**Impact of SC models on social indicators**	**ASTON**	10	33	24
T	3.1	Assess impact of covid onsupply chain and social indicators	Aston	10	16	7
T	3.2	Model for evaluating impact of new SCs on social indicators	RWI	12	30	19
T	3.3	Social accounting for sustainability	DEUSTO	14	30	17
T	3.4	The role of logistics outsourcing and impact on employment	ZLC	20	30	11
T	3.5	Analytical tools and web interface for social indicators	IRIS	18	33	16
WP	**4**	**Development of policy scenarios**	**INESC**	18	36	19
T	4.1	Identification of key horizontal issues	IML	18	24	7
T	4.2	Cross-country policy analysis	INESC	18	26	9
T	4.3	Policy scenarios building	Aston	24	30	7
T	4.4	Policy design recommendations	TUE	26	36	11
WP	**5**	**Dissemination and Communication**	**ZLC**	1	36	36
T	5.1	Dissemination and communication plan	ZLC	1	3	3
T	5.2	Stakeholder engagement and networking with experts	INESC	3	36	34
T	5.3	Dissemination and communication actions	ZLC	1	36	36

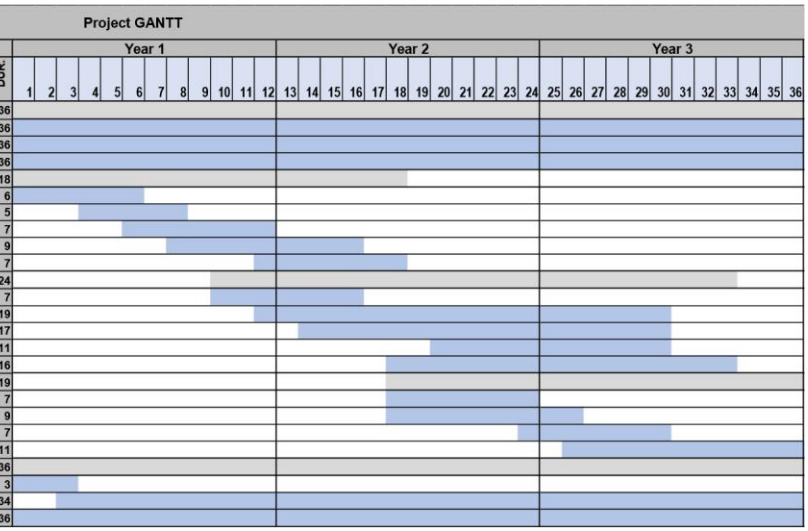

Figure 3.1 ReSChape project Gantt

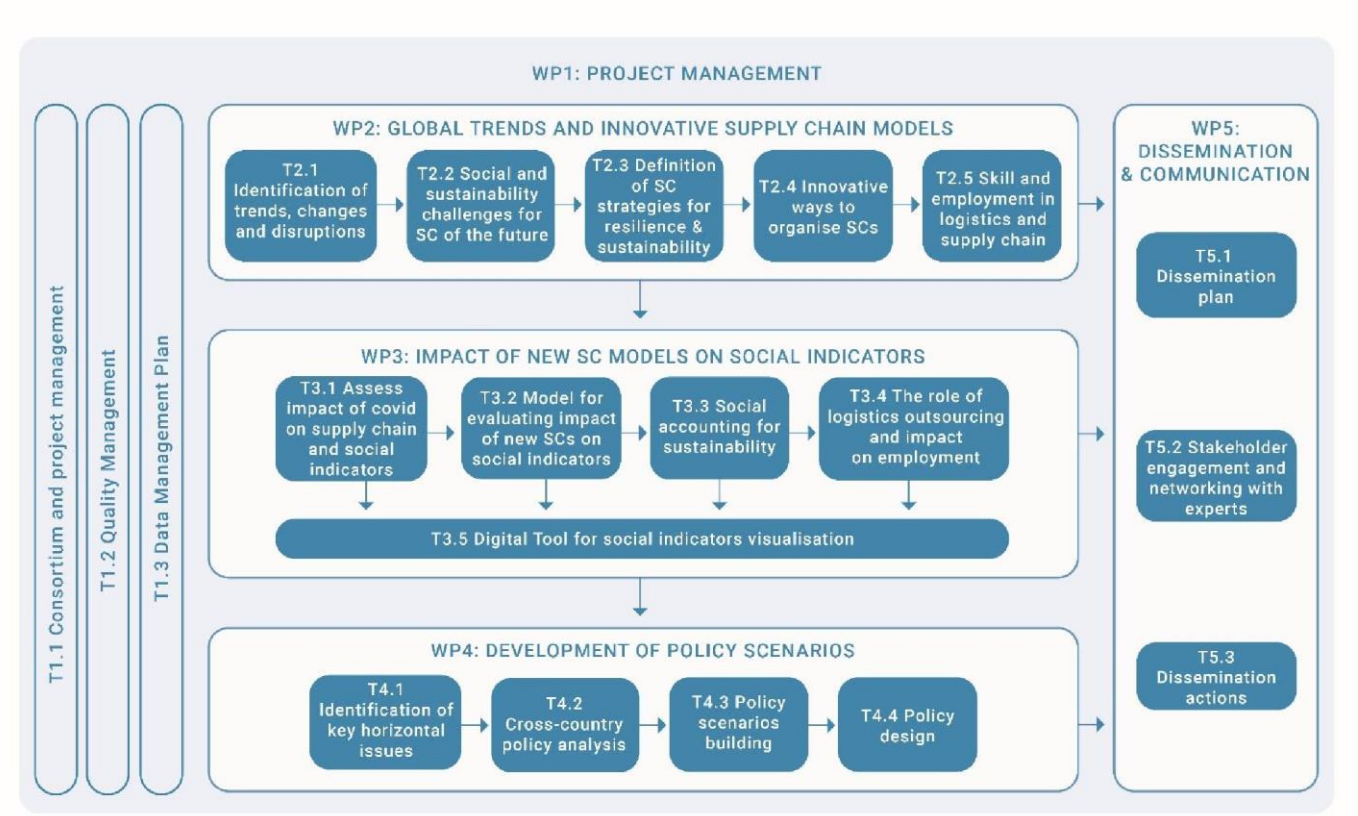

Figure 3.2 Inter-relationship between ReSChape project work packages

Table 3.1g: 'Subcontracting costs' item

No subcontracting

Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

No partners over 15% **Table 3.1i: 'Other costs categories' items (e.g. internally invoiced goods and services)** N/A **Table 3.1j: 'In-kind contributions' provided by third parties**

Updating the table 3.1j, INESC TEC decided to add IPP's (Instituto Politénico do Porto) contributions in kind. The entities UP and IPP will not be involved in the project, and no activities will be carried out by them. INESC TEC envisages the use of contributions in kind provided by UP and IPP regarding secondment of persons free of charge (Article 9.2 of the General Model Grant Agreement).

The inclusion of the IPP is due to the addition to the project team of three Professors/Researchers of this institute who have recently collaborated with INESC TEC: Hélio Cristiano Castro, Nelson Jorge Ribeiro Duarte and Cristóvão Dinis Polido Sousa. They have been developing research on the relationship between the adoption of digital technologies and sustainability in the context of supply chains and will contribute to activities related to WP2 and WP4. Professor Maria do Rosário Moreira from the Faculty of Economics of the University of Porto has researched in areas such as supply chain management, logistics and quality management, and has experience in sustainability issues in supply chains. She will mainly contribute in WP4.

Although not contemplated in the proposal phase, INESC TEC believes that the inclusion of IPP members in the team will be beneficial due to their experience in project-related topics, and because they have the potential and availability to participate in different phases throughout the entire project. On the other hand, the participation of University of Porto will be reduced compared to the original plan in the proposal, as it will be focused on WP4.

Finally, as explained in Table 3.1j, INESC TEC reinforces that "the researchers are authorized to do their research work for INESC TEC and using its research resources, such as offices and laboratories, including a clean room, meeting rooms, communication network and other supporting infrastructures. For all contractual purposes, INESC TEC assumes full responsibility for the involvement of these researchers in the project, since they are members of its research team.

Participant No. 2 - INESC TEC			
Third party name	Category	Cost (€)	Justification
Universidade do Porto (UP)	Seconded personnel	11 750 €	The University Researchers have contracts of employment with the University or Institute that signed an agreement with INESC TEC, whereby the researchers are authorized to do their research work for INESC TEC and using its research resources, like offices and laboratories, including a clean room, meeting rooms, communication network and other supporting infrastructures. For all contractual purposes, INESC TEC assumes full responsibility for the involvement of these researchers in the project, since they are members of its research team.

Instituto Politécnico do Porto (IPP)	Seconded personnel	82 250 €	Same as above

3.2 Capacity of participants and consortium as a whole

ReSChape consortium joins members from different scientific areas collecting 9 renamed organisations from 6 EU countries bringing relevant expertise and assuring connection to related communities. The consortium represents a good balance between research on industrial and technological innovation (INESC TEC, CNR), supply chains and production networks (CNR, ZLC, Aston, IML) skills and employment analysis in global trade (Aston, ZLC), models on social impact (DEUSTO, RWI, TUE), policy studies (INESC TEC, TUE) as well as software competences (IRIS).

This consortium allows to merge viewpoints that are usually studied as separate fields of research, integrating both micro-and macro-analysis on global trade, value chains, and related impact on humans bringing a perspective on their social value. This project represents a follow up of previous initiatives on digitalization, circular economy and social factors in new production models developed along the years by partners through different EU and national projects and common initiatives like INESC TEC and CNR have been working together in the FOF and ICT areas (Netchallenge, CoReNet, NextNet just to name few of them), ZLC and IML are both involved in the ALICE initiatives, CNR and IRIS are both involved in AI-CUBE with focus on role or AI also for social and ethical issues.

ReSChape consortium encompasses all the necessary expertise to achieve the project aims, and includes a wide dissemination and communication competences by all partners (in particular ZLC, WP leader), with experience in open science practices and open access publications. Gender balance is assure by the presence of several female researchers in the team which is expected to at least equal the male. This situation will be kept under control by each partner and by the coordinator. All partners in the consortium have well-defined tasks and responsibilities and their technical and non-technical expertise corresponds well to the requirements of the activities planned in the workplan.

In particular, **CNR** is the largest research organisation in Italy with more than 5000 researchers covering all science disciplines with more than 100 institutes all over Italy. In this project 2 research institutes are involved: CNR-IEIIT working on advanced research in industrial and information engineering (AI, Big Data, CPS, IoT, and others) with applications to health sector, bioengineering, manufacturing, mobility etc. to increase resilience of the whole society and supply chains; CNR-STIIMA activities aim at innovating products, processes and organisations in several research areas as: mechatronics, robotics, microsystems, AR/VR systems with focus on interaction design, new models for evaluating impact of digital technologies on organisations and humans. Both the Institutes are currently involved in several EU projects in several programmes (i.e. Health, FOF, ICT) and collaborates with several international initiatives (i.e. Vanguard, Made In Europe, KIC-ICT) and with national clusters to promote research. In particular, the Cluster of Intelligent Factories will be involved along the project with its more than 200 members from manufacturing sectors.

In this project, CNR will have the project management bringing experience from coordination of several other EU projects. In particular, Rosanna Fornasiero will be the project coordinator (previously coordinator of EU NEXTNET project-n.768884). Both the institutes will bring to the project their experience in mapping and scouting technologies, analysing social, technological and economic trends and will support the definition of new SC strategies where social innovation should meet industrial companies. The staff involved in the project has background in economics and industrial engineering and has experience in supply chain, technology and innovation management, forecasting and roadmapping for R&D policies, and management of digitalization.

INESC TEC is a private, non-profit association dedicated to scientific research and technological development, technology transfer, advanced consulting and training, and pre-incubation of new technology-based companies. INESC TEC incorporates 13 R&D Centres, structured in four thematic domains - Computer Science, Industry and Innovation, Networked Intelligent Systems, and Power and Energy and hosts over 750 integrated researchers. The Centre for Enterprise Systems Engineering (CESE) from INESC TEC will be involved in the ReSChape, with expertise on Operations Management and Enterprise Information Systems applied to industrial companies and organizational networks with application to Enterprise Cooperation Networks, Supply Chain Management; Logistic Systems; Intelligent Automation and Industry 4.0, Advanced Planning and Scheduling Systems; Big data and Business Analytics and Interoperability.

INESC TEC will be the leader of WP4, Development of policy scenarios, as well as responsible for the Definition of SC strategies for resilience and sustainability (task 2.3). For these, INESC TEC will bring his experience working with technological innovation and supply chain management, focused on Resource-efficient SCs, and on the development of policy recommendations. As member of EIT Manufacturing, EFFRA, EU Robotics, and PRODUTECH Digital Innovation, INESC TEC will actively engage industrial and technological companies and experts in the project (task 5.2).

Zaragoza Logistics Center (ZLC) is an international center of excellence for research and education in logistics and Supply Chain Management (SCM) that actively engages with industry and the public sector to develop and disseminate knowledge. ZLC was established by the Government of Aragon in Spain, in partnership with the MIT Center for Transportation and Logistics (CTL-MIT) and the University of Zaragoza. ZLC is founder and active collaborator of the European Technology Platform in Logistics, ALICE (Alliance for Logistics Innovation through Collaboration in Europe). Furthermore, ZLC has a strong network of contacts and collaborators that involves relevant European and international networks in the logistics and supply chain sector, including shippers, logistics service providers, infrastructure managers and research organizations.

ZLC will contribute to technical tasks of the project with its expertise in the supply chain domain, specifically to the definition of supply chain strategies towards sustainability and resilience, innovative ways to organize supply chains and required skills for supply chain management. Additionally ZLC will lead the task on the role of logistics outsourcing and its impact on employment. Beyond its supply chain expertise, ZLC will also contribute with its stakeholder management expertise by coordinating the dissemination and communication WP, as well as the data management task.

The RWI – Leibniz Institute for Economic Research is a leading center for scientific research and evidence-based policy advice in Germany. RWI work bridges science with practice by applying state-of-the-art methods to economic policy consulting. The overarching goals of the RWI are captured through the motto “Research with impact” along 3 channels: 1) policy impact by influencing political deliberations; 2) meeting the highest scientific standards, as evidenced by regular publications in peer-reviewed journals with a high impact factor; 3) robust impact evaluations, applied in both our consulting and research. RWI is also service providers to the scientific community through the Ruhr-Data Center (FDZ Ruhr), a publicly accessible data repository for non-commercial research. The spectrum of RWI’s work spans the core fields of economics, including labor markets, public finance, and economic growth and business cycles.

RWI is involved in ReSChape for its experience in macroeconomic models to include the impact of short-term shocks, e.g. Brexit, on variables such as value added and employment. RWI analyses longer-term developments such as the impact of the Chinese economy on the international division of labor and has built various macroeconomic models, which are constantly modified and extended. In the area of labor markets, it studies the changing quality requirements due to digitalization as well as questions of income distribution.

With a long and well established tradition, founded in 1886 as a higher education institution, DEUSTO has a mission and educational goal grounded in academic excellence and social responsibility, aiming at generating economic sustainable growth and making positive contributions towards the construction of fairer and more inclusive and human societies. Recognized as an International Excellence Campus, the university has 39 research teams working at 6 faculties (Psychology and Education, Social Sciences and Humanities, Engineering, Law, Economic and Business Sciences, and Theology). DEUSTO research is articulated around 5 interdisciplinary platforms on Social Justice and Inclusion, Ageing and Wellbeing, Creative Industries and Cities (B-Creative), Gender, Strengthening Participation, plus 4 interdisciplinary areas on Energy, Health, Territory, and Industry 4.0.

DEUSTO contributes to this proposal from its knowledge and experience in measuring the social impact of different kind of organisations. The methodology will be fitted to the global value chains context in order to analyse the social impact into both macro and micro economic indicators, wellbeing and social cohesion.

As part of the Fraunhofer Society, IML is the top choice for comprehensive logistics research and work in all fields of internal and external logistics. Teams, assembled together based on project and customer requirements, create cross-industry and customer-specific solutions in the areas of material flow technology, warehouse management, business process modelling, simulation-based business and system planning, and transport systems, resource logistics, and e-business. The department of »Supply Chain Development & Strategy« offers impulses and specific

approaches for future-oriented and strategic supply chain development. In this regard, we develop and implement tailor made business models that fit to the company's goals. Our expertise includes the conceptualization of logical structure and process design on all levels of planning, as well as the identification of economic potential and the derivation of suitable measures in the context of industry 4.0 and digitalization.

Within the project, IML will mainly work on global trends and innovative supply chain models (WP 2) as well as on the development of policy scenarios (WP 4). Leading the task 2.1, IML will identify trends, changes, and disruptions leading to social impacts in supply chain models and the European economy. Moreover, IML will work on the identification of key horizontal issues, leading task 4.1. In addition, IML will actively support the derivation of sustainable supply chain models (Task 2.2 to 2.5) as well as political fields of action (Task 4.2 and 4.3).

TUE-The department of Industrial Engineering & Innovation Sciences aims at being a European frontrunner in research and education on leading responsible and effective technology development and implementation in industry and society. TUE mission is to generate high-quality scientific progress and output, offer excellent educational programs, and create technology-based impact for society and industry, combining disciplinary and multidisciplinary excellence at the interplay of technological sciences, social sciences and the humanities. As a part of Eindhoven University of Technology, research and education in the department of Industrial Engineering & Innovation Sciences focuses on gaining knowledge on and designing solutions for complex industrial and societal challenges arising from the interplay between new key enabling technologies (e.g., robotics, artificial intelligence, green energy and hightech systems) and humans, companies and society at large.

TUE will provide expertise in the area of innovation policy design, decision analysis, and strategic roadmapping. TUE will be responsible for policy scenario analysis, and synthetizing research results into a set of policy recommendations which balance economic, environmental, and social sustainability aspects.

IRIS has broad expertise in EU-funded projects and coordination. Being a renamed engineering enterprise, IRIS designs, develops and deploys customized digital solutions and tools for industries and entities exercising activities in a broad variety of kinds, including production, processing, managing and more. IRIS mission is to accompany partners and clients thought the new Industry 4.0 paradigm, by providing flexible and adaptable solutions to the digital era such as real-time monitoring, digital storage and interfaces, back and front-end tools, etc; all of them able to be managed and monitored in association with the so called SMAC technologies environment.

IRIS will provide AI solutions development and will deliver simulations deployment to the project; as well as will be in charge of the design, development and deployment of digital analytic tools for assessing the effect in the selected policy areas.

Aston University is involved as associated partner. It is a public research university located in Birmingham, in the UK. Aston is a long-established research-led University with strong links to industry, government and commerce and strong track record in EU Funded Research. The Department of Engineering Systems & Supply Chain Management (ESSCM) in the College of Engineering and Physical Sciences has been involved also in various projects Horizon 2020, ERDF projects. ESSCM has a strong focus on building world class research capability, partnership with business and industry, and advanced learning. It comprises a multidisciplinary team of academic and research staff focused on generating and using leading edge knowledge. They have a strong focus particularly in thematic areas like systems thinking in supply chain design and integration; environmentally sustainable logistics and systems; logistics and supply chain data analytics; technology (particularly ICT) enablers; and, project risk assessment and analysis.

Aston is involved in the project for its cross-sectoral competencies especially in the field of skills and employment in logistics, supply chain sectors and the realities of supply-chain trade/ cross-border supply chains. It will be mainly involved in WP2, WP3 and WP4.

Annex 1. Affiliated partners involved in the project

<u>Affiliated party of CNR</u>

Given the framework agreement between CNR and University of Padua, registered as prot.n. 254212 on 01/07/2019, where the two parties regulate collaboration between the two institutions for research and training activities and reciprocal involvement in research projects for specific tasks, CNR envisages the implementation of some tasks by the Affiliated party University of Padua through the Department of Technologies and Industrial Management (Unipd-DTG) and Department of Economics with particular focus on the following tasks:

- WP2: Global trends and Innovative supply chain models. Unipd will be involved for supporting activities in T2.2- Social and sustainability challenges for supply chains of the future, to help classification of challenges for supply chains processes derived from social trends and in T2.3 for the definition of supply chain strategies to support resilience and sustainability applying methodologies of organizational science. Effort estimated for Unipd: 3 MM
- WP3: Impact of new SC models on social indicators. Unipd will be involved for supporting activities in T3.2 - Model for evaluating impact of new SCs on social indicators, where Unipd will support the mapping between the SC strategies and the possible social impact and definition of related indicators to be used in the models to be developed by project partners in the task. Effort estimated for Unipd: 2 MM

The total budget allocated to Unipd-DTG will be according to the following table.

Unipd-PD	Costs
Direct Personnel costs (5MM x 8.000 euro)	40.000 €
Other Direct Costs (travels)	2.000€
Indirect Costs	10.500€
Total costs	52.500 €

Short profile University of Padua

University of Padua was founded in 1222 and now provides high education to over 70,000 students, with 2,200 teachers, and support of over 2,400 administrative technicians. There are more than 13,000 graduates each year, 5,000 scholarships awarded, 32 departments, and an educational offer of excellence and in continuous transformation, the University of Padua (www.unipd.it) has long been positioned at the highest levels in the rankings of the main national and international rankings for the quality of teaching, research and services.

In particular the Department of Technologies and Industrial Management (www.gest.unipd.it) is recognized for research and training activities on engineering, technological and economic to support development of new products, materials, and production processes. The Department is part of the Engineering faculty with laboratories where researchers often work with industry in regional and European projects. In particular, the research group on operations management and supply chain management will be involved in the project with professors and researchers which have been already involved also in other European projects.

Involved Staff

Prof. Andrea Vinelli, full professor of Supply Chain and Operations Management at University of Padua

Andrea Vinelli is Full Professor at the Department of Engineering & Management - DTG - University of Padova, Italy, where he teaches Smart Operations and Supply Chain Management, Operations and Supply Chain Strategy and Service Operations Management at the Engineering & Management Graduate School. Along the years he has been delegate of the Rector in different

areas. He is author of more than 200 publications in journals such as the MIT Sloan Management Review, Corporate Social Responsibility & Environmental Management, Technological Forecasting and Social Change, International Journal of Production Economics, International Journal of Operations & Production Management, and others.

Prof. Laura Macchion, associate professor of Supply Chain Management at University of Padua.

Laura Macchion is Associate Professor at the Department of Management and Engineering of the University of Padua, where she teaches Operations Managment. Her competences are focused on Supply Chain Management and Operations Management.

Her research deals with the impact of sustainability on the management of complex and international supply networks and with the possibilities offered by new technologies for supply chain configurations. Laura Macchion has also a teaching experience in Executive and Master programs in Business Schools and is actively involved in national and international research projects.

Recent pubblications:

- Da Giau, A., Foss, N. J., Furlan, A., & Vinelli, A. (2020). Sustainable development and dynamic capabilities in the fashion industry: A multi-case study. *Corporate Social Responsibility and Environmental Management*, *27*(3), 1509-1520.
- Moretto, A., Macchion, L., Lion, A., Caniato, F., Danese, P., & Vinelli, A. (2018). Designing a roadmap towards a sustainable supply chain: A focus on the fashion industry. *Journal of cleaner production*, *193*, 169-184.
- Galeazzo, A., Furlan, A., & Vinelli, A. (2021). The role of employees' participation and managers' authority on continuous improvement and performance. *International Journal of Operations & Production Management*.
- Furlan, A., & Vinelli, A. (2018). Unpacking the coexistence between improvement and innovation in worldclass manufacturing: A dynamic capability approach. *Technological Forecasting and Social Change*, *133*, 168-178.
- Macchion, L., Marchiori, I., Vinelli, A., & Fornasiero, R. (2019). Proposing a tool for supply chain configuration: an application to customised production. In *Factories of the Future* (pp. 217-231). Springer, Cham.
- Macchion, L., Da Giau, A., Caniato, F., Caridi, M., Danese, P., Rinaldi, R., & Vinelli, A. (2018). Strategic approaches to sustainability in fashion supply chain management. *Production Planning & Control*, *29*(1), 928.
- Moretto, A., & Macchion, L. (2022). Drivers, barriers and supply chain variables influencing the adoption of the blockchain to support traceability along fashion supply chains. *Operations Management Research*, 120.

Annex II. Budget of Associated partner –Aston

The budget of Aston is represented in the table below.

A. Personnel costs	B. Sub-Contracting	C.1. Purchase Costs - Travel and Substistense	C.2. Purchase Costs - Equipments	C.3. Purchase Costs - Other goods, works and services (Consumables are here)	D. Internally Invoiced goods and services (Unit costsusual accounting practices)	Indirect costs 0.25*(A+C)	Total eligible costs
264.780 €		12.000 €		8.000 €		**71.195 €**	355.975 €